Pooling Is Not Enough: Medicaid Drug Purchasing Pools, Pharmacy-Benefit Governance, and Supplemental Rebate Capture

Abstract

Thirty-three states participate in interstate Medicaid drug purchasing pools, but whether pooling itself increases supplemental rebate capture has remained unclear because states often adopt pool membership alongside broader pharmacy-benefit governance reforms. I construct a 50-state, 2002–2024 panel linking CMS-64 supplemental rebate data to a new spell-coded pharmacy-benefit governance dataset measuring uniform preferred drug list status, pharmacy carve-out status, managed-care formulary autonomy, vendor arrangements, and rebate-negotiation models. Because the Sovereign States Drug Consortium (SSDC) is the only pool family with sufficient later adopters to support a heterogeneity-robust saturated event study, I use SSDC as the lead dynamic case and evaluate the National Medicaid Pooling Initiative (NMPI) and The Optimal PDL Solution (TOP$) as appendix extensions. In the later-adopter SSDC design, the average post-adoption effect on supplemental rebate share is 3.37 percentage points before governance controls and attenuates to 1.28 percentage points after them. A stable-design benchmark that drops reform-dense windows yields -2.05 percentage points, the standalone SSDC coefficient in a static bundle model is 1.27 percentage points ($p = 0.571$), and the standalone any-pool coefficient is essentially zero (0.06 pp, $p = 0.951$). Extensions across NMPI and TOP, switcher analyses, federal-rebate headroom checks, SDUD-based drug-mix controls, and Rambachan–Roth honest sensitivity bounds do not recover a meaningful standalone pool effect. Three related estimands organize the analysis: the *total association* between SSDC and supplemental rebate capture as implemented (3.37 pp); the *residual association* conditional on observed pharmacy-benefit governance (1.28 pp); and a *stable-design benchmark* that excludes reform-dense windows (–2.05 pp). The strongest *suggestive* positive evidence is concentrated in a bundled governance arrangement: states that simultaneously participate in a pool, maintain a uniform PDL, and carve the pharmacy benefit out of managed care show 8.91 percentage points higher supplemental rebate capture ($p = 0.005$); I treat this as a support-limited descriptive concentration result, not as the paper's headline causal finding. The pattern across the three estimands — a positive uncontrolled benchmark that attenuates sharply once governance is measured, a negative stable-design benchmark, and a near-zero standalone any-pool coefficient — is the paper's contribution: nominal pool membership is not separable from the broader governance package and is not, on its own, a lever for higher supplemental rebate capture. The paper studies rebate capture, not net drug spending; the language of "savings" is therefore reserved for explicit comparisons against net spending and is not used interchangeably with the rebate-capture outcome.

1. Introduction

1.1 The Problem: Rising Medicaid Drug Costs and the Search for Fiscal Levers

Medicaid is the single largest payer for prescription drugs in the United States. In fiscal year 2023, gross Medicaid drug spending exceeded $80 billion, with net spending after rebates reaching roughly $51 billion and continuing to rise as specialty and biologic therapies absorbed a growing share of state budgets (MACPAC, 2022). The trajectory has been relentless: net Medicaid drug spending increased 72 percent between FY2017 and FY2023, outpacing overall Medicaid spending growth and placing escalating pressure on both state and federal budgets. Managing pharmaceutical expenditures is therefore a first-order fiscal challenge for state Medicaid programs.

The urgency of that challenge is compounded by the structure of modern pharmaceutical markets. Specialty and biologic drugs account for roughly 2 percent of prescriptions but nearly 50 percent of drug spending by the early 2020s. Unlike traditional small-molecule drugs, which eventually face generic competition that drives down prices, many biologics face limited biosimilar entry and sustained pricing power. The emergence of cell and gene therapies with per-patient costs exceeding $1 million has introduced entirely new categories of fiscal risk. State Medicaid agencies must simultaneously contain growth in drug spending, maintain compliance with federal coverage rules, and preserve access for populations that rely heavily on public insurance.

State Medicaid programs operate within a mixed federal-state structure that shapes their cost-containment options. At the federal level, the Medicaid Drug Rebate Program (MDRP), established by the Omnibus Budget Reconciliation Act of 1990, requires manufacturers to pay statutory rebates in exchange for Medicaid coverage. Those mandatory rebates create the floor for Medicaid drug pricing. At the state level, however, agencies retain meaningful discretion over several additional levers: preferred drug list (PDL) design, prior authorization rules, managed-care pharmacy architecture, drug utilization review, and supplemental rebate negotiations. This chapter focuses on the last of those levers.

1.2 Supplemental Rebates: A Critical and Understudied Fiscal Lever

Supplemental rebates are voluntary rebates negotiated above and beyond the federal MDRP minimum. Manufacturers pay them in exchange for preferred formulary placement, easier utilization management, or broader access to a state's covered population. As of 2019, 47 states and the District of Columbia had supplemental rebate agreements, collectively generating over $1.7 billion annually (OIG, 2014; KFF/HMA, 2020). Manufacturers pay these rebates to ensure preferred placement on the state's PDL, which reduces prior authorization requirements and other utilization management barriers.

Despite that fiscal importance, supplemental rebates have received remarkably little rigorous empirical attention. The Office of Inspector General documented substantial cross-state variation in how states collect and structure supplemental rebates but did not attempt to explain it (OIG, 2014). MACPAC (2018) estimated that supplemental rebates account for roughly 6 percent of total Medicaid drug rebates but did not investigate the drivers of variation. The academic Medicaid pharmacy literature has focused on adjacent policies—managed care versus fee-for-service drug spending, formulary design, prior authorization, generic substitution—leaving supplemental rebates as one of the largest policy-relevant blind spots in the evidence base.

That omission matters because the cross-state variation is large. In 2024, supplemental rebates ranged from near zero in some states to over 28 percent of drug spending in others. If that variation reflects replicable institutional arrangements, the policy implications are substantial. If it instead reflects idiosyncratic characteristics or bundled reforms that cannot be transferred, the lesson is very different. Distinguishing between these explanations requires causal evidence, which to date does not exist.

1.3 Interstate Drug Purchasing Pools: Widespread Adoption, Little Causal Evidence

Many states attempt to increase bargaining leverage by joining interstate drug purchasing pools. The intuitive logic is straightforward: a larger covered population should strengthen the threat to steer utilization toward preferred products, which should in turn raise supplemental rebates. Three major pool families emerged in the mid-2000s: the National Medicaid Pooling Initiative (NMPI, established 2003), The Optimal PDL Solution (TOP$, established 2005), and the Sovereign States Drug Consortium (SSDC, established 2005). By 2024, 33 of 50 states participated in one of those three arrangements.

That widespread adoption is theoretically plausible. Buyer-side market power models suggest that larger, more coordinated purchasers can secure better prices when they can credibly redirect demand (Chown, Dranove, Garthwaite, and Keener, 2019). Structural models of formulary-rebate contracting imply that rebates depend on covered lives and on the buyer's ability to commit to preferred placement (Ho and Lee, 2024). International pooled-procurement research reaches a similar conclusion: pooling can improve procurement outcomes, but the size of the gain depends heavily on governance and implementation (Dubois, Lefouili, and Straub, 2021; Parmaksiz et al., 2022).

Yet despite those strong priors, there is no settled causal evidence on whether interstate Medicaid purchasing pools actually increase supplemental rebate capture. Existing accounts are entirely descriptive: Horvath (2019) documented that Vermont reported 4.7 percent additional savings in its first year of SSDC participation and that New York reported $80.5 million in additional savings through NMPI, but these figures lack counterfactual comparisons. In short, two-thirds of U.S. states have adopted a policy intervention for which the evidence base consists entirely of anecdotes and self-reported savings figures.

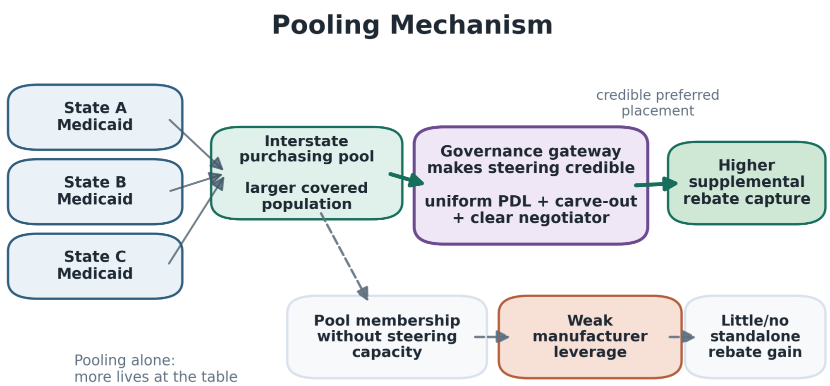

Figure 1: Pooling mechanism

1.4 This Chapter's Contribution

This chapter makes three contributions. First, it constructs a novel 50-state panel of Medicaid drug-pool membership and pharmacy-benefit governance from 2002 through 2024, linking the CMS-64 rebate panel to a new spell-coded pharmacy-benefit structure dataset that directly measures uniform preferred drug list status, pharmacy carve-outs, managed-care formulary autonomy, vendor arrangements, rebate-negotiation model, and major reform episodes. Second, it provides quasi-experimental evidence on whether interstate purchasing pools increase supplemental rebate capture once these governance features are measured directly, organized around three explicit estimands: the *total association* between SSDC adoption and supplemental rebate capture as implemented (the policy-bundle estimand); the *residual association* conditional on observed pharmacy-benefit governance (closer to a controlled-direct-effect estimand); and a *stable-design benchmark* that excludes reform-dense windows. Third, it shows that nominal pool membership is a weak predictor of rebate performance once those three estimands are read together; observed gains around pool adoption are not separable from the broader governance package, and any positive bundle performance concentrates in states that combine pool participation with centralized formulary and pharmacy-benefit control. The contribution is the *attenuation pattern* across the three estimands, not any single coefficient.

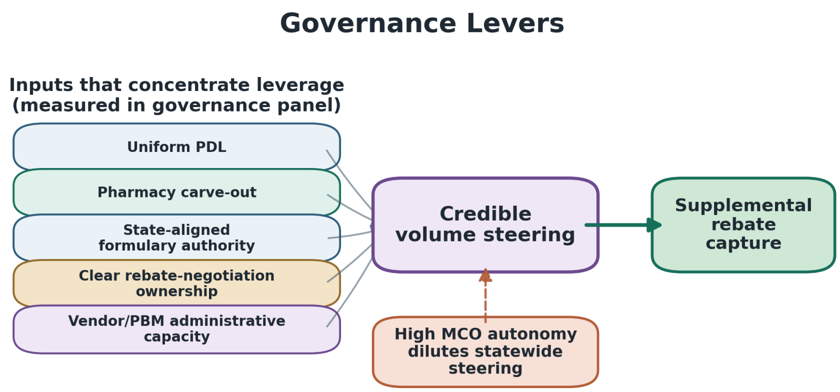

Figure 2: Governance levers

The core identification problem motivating this approach is not simply that some states are permanently different from others. It is that states often change multiple pharmacy-benefit features at roughly the same time they join a pool. If a state adopts a uniform preferred drug list, changes who negotiates rebates, or carves pharmacy out of managed care near SSDC entry, a positive SSDC coefficient in a model with only state and year fixed effects can still reflect those other reforms. State fixed effects absorb time-invariant differences across states, and year fixed effects absorb common shocks, but neither absorbs within-state, time-varying changes in PDL governance, carve-out status, or vendor arrangements that occur near SSDC entry. By measuring time-varying governance changes directly, the design asks a sharper question: does interstate pooling still appear to increase supplemental rebate capture once the main concurrent reforms are accounted for, and where in the governance landscape does observed rebate capture actually concentrate?

I use SSDC as the lead dynamic case because it is the only pool family with enough later adopters to support an informative heterogeneity-robust saturated event study. NMPI and TOP are evaluated through appendix extensions, static pool-family interaction models, a switcher subanalysis, outcome-construction checks, and an exploratory pool-size heterogeneity branch. A complementary any-pool specification with two- and three-way governance interactions tests whether the standalone pool result generalizes beyond SSDC and where any positive bundle effect concentrates.

The remainder of the chapter proceeds as follows. Section 2 reviews institutional background and the relevant literatures. Section 3 describes the data. Section 4 presents the empirical strategy. Section 5 reports results. Section 6 discusses implications and limitations. Section 7 concludes.

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2. Background and Literature Review

2.1 The Medicaid Drug Rebate Program

A short primer on Medicaid pharmacy benefits. Medicaid is the single largest payer for prescription drugs in the United States, but each state's Medicaid pharmacy benefit operates within a layered federal-state architecture that determines how drug spending and rebates flow. At the federal level, the Medicaid Drug Rebate Program (MDRP) sets a statutory rebate floor — manufacturers must rebate a minimum share of every Medicaid prescription back to the federal government in exchange for Medicaid coverage of their products. At the state level, state Medicaid agencies decide which drugs to make easier or harder for clinicians to prescribe, primarily through the **Preferred Drug List (PDL)** — a state-published list of which drugs are "preferred" (no prior authorization, easier prescribing) and which are "non-preferred" (requires prior authorization, sometimes a step-therapy fail-first requirement). Manufacturers compete for preferred placement on the PDL by offering **supplemental rebates** above the federal MDRP minimum; preferred placement materially increases a drug's Medicaid market share within its therapeutic class, so manufacturers are willing to pay supplemental rebates to capture it. The state's bargaining leverage in supplemental-rebate negotiations is, mechanically, a function of how many covered lives it can credibly steer with a PDL decision.

That bargaining leverage is shaped by two further architectural choices. **Pharmacy benefit carve-in vs. carve-out** determines whether the Medicaid pharmacy benefit is administered by the state directly (carve-out — the state controls the PDL and rebates for the entire pharmacy benefit, including for managed-care enrollees) or by managed-care organizations on contract with the state (carve-in — MCOs run their own formularies for their enrolled members, with statewide coordination varying widely). **Uniform PDL** policies require all MCOs to use the state's PDL rather than negotiate their own — converting a fragmented set of MCO formularies back into a single statewide formulary decision. A state can be in any of four corners of the (uniform PDL $\times$ carve-out) space, and the size of its credible bargaining unit varies accordingly. **Interstate purchasing pools** — the policy lever this paper studies — sit on top of all of this: they combine multiple states' covered lives into a single negotiating unit. Whether pool membership translates into higher supplemental rebate capture should depend on the state's ability to translate pooled covered lives into a coordinated PDL decision, which in turn depends on the carve-out and uniform-PDL architecture.

The MDRP statute. The Medicaid Drug Rebate Program (MDRP) was established by the Omnibus Budget Reconciliation Act of 1990 and has been a central feature of Medicaid pharmaceutical policy for more than three decades.

Manufacturers must enter into rebate agreements with the Secretary of Health and Human Services as a condition of Medicaid coverage. For brand-name drugs, the federal unit rebate amount is the greater of 23.1 percent of Average Manufacturer Price (AMP) or the difference between AMP and the manufacturer's best price—the lowest price offered to any commercial purchaser, with certain exclusions—plus an inflation penalty when price growth exceeds CPI-U. For generics, the statutory minimum is 13 percent of AMP.

This pricing structure creates well-documented incentive distortions. Duggan and Scott Morton (2006) provided foundational evidence, demonstrating that manufacturers respond to the best-price provision by strategically raising prices to commercial purchasers. Their estimates indicate that a 10-percentage-point increase in Medicaid market share is associated with 7–10 percent higher average drug prices across all purchasers. This cross-subsidization means that the MDRP's mandatory rebates are partially offset by higher commercial prices.

The best-price provision has additional consequences for supplemental rebate negotiations. Because manufacturers must offer Medicaid a rebate at least as large as the AMP-best-price spread, any price concessions to large commercial purchasers automatically increase the statutory rebate. This creates a floor below which supplemental negotiations operate: the supplemental rebate is an increment above an already-deep statutory discount, making the marginal dollar obtained through supplemental negotiation more difficult to extract.

The Affordable Care Act intensified the MDRP's role in 2010 through two changes relevant to this analysis. First, it raised the minimum brand-name rebate from 15.1 to 23.1 percent of AMP, substantially raising the statutory floor. Second, it extended mandatory MDRP rebates to drugs dispensed through Medicaid managed care, closing a loophole that had allowed MCOs to negotiate outside the MDRP framework. After 2010, supplemental rebates could be layered on top of statutory rebates for both fee-for-service and MCO-dispensed drugs, increasing the potential scope of supplemental rebate programs but also adding administrative complexity. That policy shift is one reason the modern post-2010 window is analytically important.

2.2 Supplemental Rebates and Preferred Drug Lists

Supplemental rebates are voluntary payments that manufacturers make in exchange for preferred formulary placement. They are embedded in formulary design, prior authorization policy, and the state's ability to steer utilization toward preferred products.

The economics of supplemental rebates are best understood through the lens of formulary contracting theory. Ho and Lee (2024) developed a structural model of formulary-rebate contracting between PBMs and manufacturers, showing that multi-tier formularies substantially increase negotiated rebate payments. In their model, rebate contracts specify per-unit payments contingent on formulary placement and competitor positioning. A manufacturer whose drug is placed on the preferred tier gains market share within its therapeutic class, while the payer captures a rebate that partly offsets coverage costs. The size of the rebate depends on the manufacturer's outside option (losing preferred status) and the payer's outside option (switching to an alternative). That logic implies that supplemental rebate negotiation is strongest in therapeutic classes with credible alternatives and weakest in sole-source or minimally substitutable products.

Chown, Dranove, Garthwaite, and Keener (2019) frame this intuition in their analysis of buyer-side market power in health care. Comparing the United States and Canada, they argue that prescription-drug price differences more directly reflect public buyer power than do provider-wage differences. Their analysis supports the broader premise that purchasing scale matters only when the buyer can translate that scale into credible demand control.

Preferred drug lists are the operational mechanism through which that leverage is exercised. Virabhak and Shinogle (2005) documented large prescribing responses to PDL implementation: restricted-drug prescription shares fell by 9.0 percentage points in Illinois Medicaid and 6.2 percentage points in Louisiana Medicaid after PDL adoption. Munshi et al. (2018) found that Florida's state-mandated PDL reduced overall drug use by 6 percent but increased plan costs by 27 percent—an apparently paradoxical result attributable to the PDL's channeling of utilization toward preferred branded drugs with high acquisition costs but also high rebate yields. This finding illustrates that PDL effects on spending are not straightforward: a PDL that maximizes supplemental rebate revenue may not minimize net drug costs if it channels utilization away from lower-cost generics.

States vary substantially in their PDL architecture. Some maintain a uniform PDL spanning both fee-for-service and managed-care enrollees, giving the state direct control over formulary placement for its entire Medicaid population. Others allow MCOs to maintain their own formularies, fragmenting the state's bargaining leverage by splitting the covered population across multiple formulary decisions. A state with a uniform PDL can credibly commit to placement decisions for a larger share of its covered lives, which should increase bargaining power. Conversely, a state may join a pool but still fail to convert pooled covered lives into real negotiating power if managed care plans retain substantial formulary autonomy.

2.3 Interstate Drug Purchasing Pools

Three major pool families emerged in the mid-2000s, and they differ in ways that matter for identification and interpretation.

NMPI negotiates collective supplemental rebate agreements (SRAs) on behalf of member states. The pool entity contracts directly with manufacturers, while member states choose which negotiated drugs to place on their own formularies. The pool—not the individual state—owns the contract paper. This structure offers efficiency advantages (a single negotiation covers multiple states) but may reduce state-level customization. By 2024, 11 states participated, including several large states such as New York and Michigan.

**TOP$** negotiates collective SRAs similarly to NMPI. Limited publicly available information exists on its specific contracting approach; its structure is assumed to be broadly similar to NMPI based on available documentation. By 2024, 7 states participated (Connecticut, Idaho, Louisiana, Maryland, Nebraska, Tennessee, Washington). The relative obscurity of TOP$'s operations reflects a broader challenge in studying Medicaid purchasing pools: much of the relevant institutional detail is not publicly documented.

SSDC is structurally distinct from the other two pools in a critical way: member states contract *individually* with manufacturers using their own state-specific SRAs, while leveraging the collective purchasing power of the consortium's combined covered lives. Each state retains full ownership of its contracts and complete autonomy over its PDL. The consortium provides a negotiating framework, shared market intelligence, and the leverage of its combined membership, but the actual contract execution is between the individual state and the manufacturer. By 2024, 15 states participated.

This structural distinction—individual contracting with pooled leverage, as opposed to collective contracting with individual PDL selection—may explain SSDC's apparently differential descriptive performance. Several features of the SSDC model may contribute: (1) state PDL autonomy, allowing formulary placement tailored to the population's clinical needs and class-specific competitive dynamics; (2) direct contract ownership, aligning incentives so the state bears the full cost of weak negotiation and captures the full benefit of strong negotiation, avoiding free-rider problems inherent in collective models; (3) customized negotiating positions based on the state's unique market characteristics; and (4) coordinated market intelligence from the consortium while preserving decentralized execution.

By 2024, the observed landscape included 15 SSDC states, 11 NMPI states, 7 TOP states, and 17 states in no major pool. The panel also reveals five states that switch from NMPI or TOP into SSDC over time, suggesting states did not treat the pools as perfect substitutes. A model that treats every pool as interchangeable implicitly assumes that collective contracts and individual state-specific contracts are close substitutes once covered lives are large enough. That assumption is not obviously right.

2.4 Pharmacy-Benefit Structure as a Confounder

The central analytical logic of this chapter is that pharmacy-benefit structure is not peripheral context—it is the main competing explanation for an apparent pool effect. States may join a pool while also adopting a uniform preferred drug list, changing whether the pharmacy benefit is carved in or out of managed care, changing vendor arrangements, altering plan formulary autonomy, or shifting the rebate-negotiation model itself.

Each of those changes can plausibly affect supplemental rebate capture. A uniform preferred drug list may strengthen statewide leverage even without any pool change. A carve-out can centralize control over a larger share of covered lives. Greater managed-care formulary autonomy can weaken the state's ability to deliver on preferred placement. A different rebate-negotiation model may matter more than nominal pool membership. If those changes occur near pool entry, a simple SSDC coefficient in a state-year fixed-effects model can still be badly confounded.

The interaction between pharmacy benefit design and supplemental rebate capture is well grounded in the existing literature. Dranove, Ody, and Starc (2021) provide the most rigorous causal study of Medicaid pharmacy benefit privatization, finding that full privatization decreased drug spending by 22.4 percent, with two-thirds of savings from greater generic use and one-third from lower branded drug prices. Crucially, effects were concentrated in states giving MCOs formulary flexibility—implying that MCO formulary autonomy can either complement or undermine pool-negotiated PDLs. Bendicksen and Kesselheim (2022) documented a countervailing trend of pharmacy benefit carve-outs, noting that 8 states had carved out by 2023; California estimated $859 million in annual savings. Hernandez and Gellad (2020) highlighted that fee-for-service programs tend to retain expensive branded drugs to capture large inflation rebates under the MDRP formula, while MCOs adopt generics at much higher rates. This dynamic is relevant because pool-negotiated PDLs may favor branded drugs with large supplemental rebates over lower-cost generics, creating tradeoffs between supplemental rebate maximization and overall cost minimization.

2.5 Theoretical Foundations and International Evidence

The theoretical literature provides strong priors for why pooling might matter while also making clear why pooling alone need not be sufficient. Chown et al. (2019) argue that buyer-side market power can lower prices when purchasers can credibly redirect demand. Ho and Lee (2024) show that covered lives matter only insofar as they are attached to a credible formulary decision. Both frameworks imply that bargaining leverage depends on governance and implementation, not simply on nominal pool size.

International pooled-procurement evidence reaches a similar conclusion. Dubois, Lefouili, and Straub (2021) study centralized procurement across seven low- and middle-income countries and find that pooled public procurement is associated with lower drug prices, with smaller reductions when the supply side is more concentrated. The implication is directly relevant here: scale matters less when sellers face weak competitive pressure or when the buyer cannot credibly move volume. Parmaksiz et al. (2022), in a systematic review of 44 empirical studies on pooled procurement, similarly emphasize the capacities and alignment needed among buyers, pooled-procurement organizations, and suppliers. Savings ranged widely, suggesting that pool design features—not pooling per se—drive outcomes.

Those lessons map naturally onto the Medicaid setting. If state-level administrative architecture determines whether a pool can actually deliver preferred placement, then the effect of joining a pool should depend heavily on uniform PDLs, carve-outs, formulary governance, and negotiation structure.

2.6 Staggered Difference-in-Differences Methodology

I follow the heterogeneity-robust staggered-DiD literature. The lead dynamic specification is a saturated later-adopter event study; the primary cross-check is the Callaway and Sant'Anna (2021) group-time ATT estimator using not-yet-treated controls. Two-way fixed effects is reported only as a benchmark and diagnosed via the Goodman-Bacon (2021) decomposition. Sensitivity to deviations from smooth parallel trends is assessed using the Rambachan and Roth (2023) honest-bounds framework, with the breakdown ratio M/M as the interpretively sharp summary. SSDC adoption is staggered over nearly two decades — six early adopters in 2006–2007 and nine later adopters from 2011 onward — and this design choice (later-adopter saturated event study, with not-yet-treated comparators) is what makes the standard heterogeneity-bias diagnostics interpretable in a sample with this much within-period variation.

2.7 Why Effects May Arrive Slowly Even When Real

Even if interstate pooling were genuinely effective, a large same-year discontinuity in supplemental rebate capture would be unlikely. States do not instantaneously renegotiate every therapeutic class the moment they enter a pool. Supplemental rebates are implemented through contract renewals, PDL reviews, prior authorization updates, and class-by-class negotiations with manufacturers—implying a ramp-up process spanning one to three years.

That institutional timing has two implications. First, a front-loaded null does not by itself refute pooling. Second, a gradual pattern is not uniquely diagnostic of a clean pool effect, because the same graduality could also arise from a broader administrative package implemented over time. The paper therefore cannot point to a gradual path and declare victory for the pool-only story; the design is built to distinguish those possibilities.

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3. Data

This section describes the construction of the dataset, which links the original rebate panel to a new 50-state pharmacy-benefit panel and to SDUD-based drug-mix measures.

3.1 CMS-64 Financial Management Reports

The primary outcome is annual supplemental rebate capture, defined as:

Supplemental Rebate Share = Supplemental Rebate Dollars / Total Drug Expenditures

Both numerator and denominator come from CMS-64 Financial Management Reports. I use a share rather than a dollar outcome because states differ enormously in program size; the share-based outcome asks whether a state is extracting larger supplemental rebates relative to the scale of its own drug program.

Constructing a consistent CMS-64 panel requires harmonization across multiple reporting formats and line-item conventions. The raw data files span multiple formats: one master file covers fiscal years 2002–2011, while individual annual files cover later years. I compiled all years into a master file with standardized variable names and consistent state identifiers. A total of 39 state-year observations (3.4 percent of the panel) have missing supplemental rebate data, left as missing rather than imputed. Six state-years exhibit negative supplemental rebate values, which is conceptually difficult to reconcile with voluntary rebate payments; four are negligible in magnitude, while North Carolina in 2013 (-8.1 percent) and Virginia in 2012 (-3.0 percent) are larger and likely reflect timing or reporting artifacts. These observations are retained without adjustment; excluding them does not materially affect results.

After standardization, the rebate panel contains 1,150 state-years from 2002 through 2024, with 1,111 nonmissing supplemental rebate observations. The CMS-64 reports on a fiscal-year basis, which may not align perfectly with calendar-year pool membership indicators. Because pool membership changes typically take effect at the beginning of a state fiscal year and supplemental rebate agreements cover annual or multi-year periods, this misalignment is unlikely to introduce systematic bias.

3.2 State Drug Utilization Data

State Drug Utilization Data (SDUD) enter the chapter in two ways. First, they serve as an external validation source for the CMS-64 outcome. Total drug spending figures are cross-referenced against CMS-64 to identify states where the two sources diverge substantially. Discrepancies are expected due to differences in reporting scope (SDUD covers outpatient pharmacy only; CMS-64 may include physician-administered drugs), timing (calendar-year versus fiscal-year), and varying inclusion of managed care drug spending.

Second, in the appendix analyses, SDUD provides state-year drug-mix controls intended to address the critique that supplemental rebate percentages may partly reflect what kinds of drugs states use rather than stronger negotiating leverage. The SDUD mix panel summarizes each state-year into several descriptive measures: single-source spend share, innovator-multiple-source spend share, average spend per prescription, NDC-level concentration, and the share of spending accounted for by the top ten NDCs. These measures are sufficient to test whether simple differences in drug baskets drive the main results. The category crosswalk's matched spending share rises from 26.6 percent in 2002 to

99.5 percent in 2024, with an average of 79.4 percent across 2010–2024, so these measures are most informative in the later part of the panel.

3.3 Pool Membership Data

The first policy panel documents pool membership for all 50 states from 2002 through 2024. No pre-existing public dataset provides a fully reconciled state-year history of SSDC, NMPI, TOP, and no-pool participation. I constructed this dataset through three complementary approaches: systematic scraping of state Medicaid agency websites including archived historical content accessed through the Wayback Machine; direct contact with each state's MDRP contacts via phone and email; and FOIA requests or state Public Records Requests for states with incomplete responses. The resulting timeline was validated against KFF State Health Facts (2019), NASHP (Horvath, 2019), OIG (2014), and expert interviews.

Pool indicators are mutually exclusive within a state-year. Each state-year is coded as SSDC, NMPI, TOP, or NoPool. Five states switch from NMPI or TOP into SSDC during the panel: Delaware, Kentucky, Pennsylvania, Vermont, and West Virginia.

3.4 Pharmacy-Benefit Structure Panel

The chapter's key empirical contribution is a second state-year policy panel that measures pharmacy-benefit structure directly. The panel contains 747 coded state-years across all 50 states and tracks seven features: uniform PDL status, pharmacy carve-out status, MCO formulary autonomy, PDL vendor, PBM vendor, rebate-negotiation model, and major reform flags.

These variables are the minimum design features needed to test the bundled-confounding story. If the apparent pool effect attenuates sharply after conditioning on them, the positive benchmark is more plausibly a bundle effect than a clean pool effect.

The coding strategy is intentionally conservative. Many spell boundaries are supported by official current manuals, archived state materials, and managed-care documents, but some spells remain medium-certainty backcasts from later official manuals rather than fully contemporaneous archived sources. I built the panel at the spell level rather than coding every year from scratch. Each spell records a start year, end year, source basis, certainty rating, and coder notes. That structure makes the stable-design rule transparent: state-years near known transitions can be dropped even when the exact monthly timing is uncertain.

3.5 Treatment Timing and Analytic Samples

SSDC adoption timing is central to the design. The fifteen SSDC adopters enter in ten cohorts:

Cohort	States
2006	Iowa, Maine, Vermont
2007	Oregon, Utah, Wyoming
2011	North Dakota
2014	Mississippi
2015	West Virginia
2016	Delaware, Oklahoma
2017	Ohio
2020	Pennsylvania
2022	South Dakota
2024	Kentucky

The six early adopters in 2006 and 2007 are already treated at the start of the modern post-2010 window, so they cannot contribute untreated pre-period support in a later-adopter event study. The main dynamic sample consequently focuses on the nine later adopters first treated in 2011 or later, yielding 660 observations across 44 states in the uncontrolled specification and 649 once pharmacy-benefit controls are required. The stable-design benchmark, which excludes state-years within plus/minus two years of major benefit transitions, retains 586 observations across all 50 states.

Early adopters (2006–2007) were predominantly smaller, rural states with limited internal pharmaceutical negotiating capacity. The 2011–2017 wave brought a mix of small and medium-sized states, including Ohio. The most recent adopters include Pennsylvania, one of the largest state Medicaid programs. This variation in adopter characteristics is important for interpreting generalizability.

3.6 Descriptive Statistics

The merged dataset contains 1,150 state-years, with 1,111 nonmissing supplemental rebate observations and complete key governance controls for 64.96 percent of the merged file.

Table 1. Descriptive Statistics by Pool Family

Pool family	Mean supp. rebate share	Median	SD	Min	Max	State-years
NoPool	1.93%	0.26%	2.92%	-2.98%	21.83%	555
SSDC	5.15%	4.72%	4.38%	0.00%	28.40%	180
NMPI	2.83%	1.99%	3.39%	-8.11%	28.24%	219
TOP	3.33%	3.52%	2.49%	-0.08%	12.35%	157

SSDC state-years exhibit substantially higher supplemental rebate percentages than all other categories. The gap between SSDC and no-pool states is approximately 3.2 percentage points at the mean and 4.5 points at the median. NMPI and TOP state-years are closer to non-pool states. The large gap between mean and median for no-pool states (1.93 vs. 0.26 percent) reflects a right-skewed distribution with many states reporting near-zero supplemental rebates. But SSDC state-years also differ in governance and utilization profiles—exactly why the analysis conditions on those features directly.

Table 2. Coverage and Governance Support

Metric	Value
Rebate panel state-years	1,150
State-years with nonmissing supplemental rebate outcome	1,111
Coded state-years in pharmacy-benefit panel	747
States covered in pharmacy-benefit panel	50
SSDC adopters coded	15

Table 2b. Pharmacy-Benefit Governance Panel — Per-Variable Coverage

Variable	State-years coded	States coded	Non-missing share
Uniform PDL status	747	50	100.0%
Pharmacy carve-out status	747	50	100.0%
MCO formulary autonomy	747	50	100.0%
Rebate-negotiation model	747	50	100.0%
Major reform flag	747	50	100.0%
PDL vendor identity	14	4	1.9%
PBM vendor identity	39	5	5.2%

The five variables that enter the controlled and bundle specifications are fully coded across all 747 state-year spells; vendor identity is sparser and enters only as appendix descriptive material rather than as a control.

Table 2c. Coding-Certainty Composition by Treated Status

State group	High-certainty spells	Medium-certainty spells	High-certainty share
SSDC adopters (state-years)	39	193	16.8%
Never-treated states (state-years)	41	474	8.0%

A higher share of treated-state spells is documented as high-certainty than never-treated-state spells, reflecting that SSDC adopters tend to publish more recent, more accessible pharmacy-benefit documentation than smaller never-treated states. The dependence of the controlled estimates on medium-certainty backcasts in never-treated states is a remaining limitation flagged in §6.5.

Table 2d. Sample Selection: Complete vs Incomplete Governance Controls

Characteristic	Complete controls	Incomplete controls
State-years	717	433
States	49	49
Mean supplemental rebate share	3.35%	1.89%
SSDC share	22.7%	4.2%
Any-pool share	61.5%	31.6%
Post-2010 share	98.9%	9.5%

The complete-controls sample is concentrated in the post-2010 window where the governance panel is dense; the incomplete-controls sample is overwhelmingly pre-2010 state-years, when the rebate panel exists but the pharmacy-benefit spell coding is thinner. This is a meaningful selection feature: the controlled estimate is identified almost entirely on the modern post-MDRP window. I therefore interpret the controlled estimates as descriptive of the post-2010 governance landscape rather than the full 2002–2024 panel. | Never-treated states coded | 35 | | Share of merged file with complete key controls | 64.96% |

Table 2a. Drug-Mix Descriptives by Pool Family (SDUD)

Pool family	Single-source spend share	Innovator multi-source share	Avg. spend/Rx	Top-10 NDC share
NoPool	24.5%	24.2%	$79.72	14.2%
SSDC	45.5%	23.3%	$91.46	18.7%
NMPI	34.6%	23.9%	$88.48	16.4%
TOP	35.2%	27.1%	$88.57	15.8%

These drug-mix differences confirm that state drug baskets vary across pool families, motivating the SDUD-based composition checks in the appendix.

4. Empirical Strategy

4.1 Identification

The causal question is whether interstate pool participation increases supplemental rebate capture. In the lead design, the treatment is SSDC adoption. The identification challenge extends beyond selection into SSDC: because SSDC adoption can coincide with other within-state pharmacy reforms that also affect supplemental rebate capture, a positive SSDC coefficient in a model with only state and year fixed effects is insufficient. State fixed effects absorb time-invariant differences; year fixed effects absorb common shocks. Neither absorbs within-state, time-varying changes in PDL governance, carve-out status, rebate-negotiation structure, or vendor arrangements that occur near SSDC entry.

The parallel-trends requirement is therefore not the only identification hurdle. Even if untreated states provide a plausible trend counterfactual, the policy contrast itself can be mismeasured if "SSDC adoption" implicitly bundles changes in how the state governs the pharmacy benefit. The design addresses both problems by adding time-varying pharmacy-benefit controls. The paper's most policy-relevant comparison is consequently the difference between the uncontrolled and controlled versions of the same design.

Three estimands. Because the time-varying governance variables can play either a confounding role or a mediating role—or both—it is important to be explicit about what each specification recovers. I distinguish three estimands. The uncontrolled later-adopter event study estimates the *total association* between SSDC adoption and supplemental rebate capture as SSDC was implemented in adopting states; this is the policy-bundle estimand and includes whatever pharmacy-benefit reforms moved together with pool entry. The governance-controlled model estimates the *residual association* with SSDC after conditioning on time-varying pharmacy-benefit architecture; this is closer to a pool-only or controlled-direct-effect estimand, but it may also control away mechanisms if governance reforms are part of how SSDC is implemented in practice. The stable-design benchmark, which excludes state-years near major benefit transitions, asks whether a residual pool effect remains when reform-dense windows are removed at the cost of selection on policy stability. I therefore interpret attenuation after governance controls not as proof that SSDC had no effect, but as evidence that the observed gains are not separable from the broader governance package. The any-pool triple-interaction model in Section 5.14 supplements these estimands by asking where in the governance landscape any positive bundle effect actually concentrates.

4.2 Main Later-Adopter Saturated Event Study

The main dynamic specification is a later-adopter heterogeneity-robust saturated event study. The estimating equation is:

Y*_{st}* = *_s* + *_t* + Σ*_k* *_k* · 1[event_time*_{st}*

= k] · 1[later_treated*_s*] + X{st}' + *{st}*

where Y*_{st}* is supplemental rebate share for state *s* in year *t*; *_s *and* _t* are state and year fixed effects; event time *k* = *t* - *g_s*, where *g_s* is the first year of SSDC participation; *k* = -1 is the omitted reference period; X*_{st}* is the vector of time-varying pharmacy-benefit controls (in the controlled specification); and standard errors are clustered at the state level.

I use only the nine SSDC states first treated in 2011 or later together with valid untreated comparison support. This design avoids relying on the six early adopters that are already treated by 2010 and therefore cannot contribute untreated pre-period support. The saturated structure avoids the contamination problems that Goodman-Bacon (2021) and de Chaisemartin and D'Haultfoeuille (2020) documented for standard TWFE.

I report the average post-adoption effect over event times 0 through 10 as the key summary parameter, rather than emphasizing any single lag, because later lags are estimated from fewer treated states and early lags need not capture the full effect even when the treatment is meaningful.

4.3 Governance-Controlled Design and Stable Benchmark

I then re-estimate the later-adopter saturated event study after adding the time-varying pharmacy-benefit controls from the spell-coded panel. This is the chapter's core specification: it asks whether the SSDC benchmark survives once the main bundled confounders are measured directly.

I also estimate a stable-design linear benchmark that excludes state-years within a plus/minus two-year window of major pharmacy-benefit transitions. Major transitions include changes in uniform PDL status, carve-out or carve-in switches, changes in PDL or PBM vendor, changes in the rebate negotiation model, and explicit major reform flags. The plus/minus two-year exclusion window is deliberately conservative because governance transitions often unfold over adjacent fiscal years. This benchmark provides a directional check on whether the positive result persists once reform-heavy windows are removed.

4.4 Static Bundle Model and Broader Pool Extensions

The static bundle model estimates supplemental rebate capture as a function of SSDC adoption, uniform PDL status, pharmacy carve-out status, and SSDC-by-governance interactions, together with state and year fixed effects. It cannot recover the full timing path, but it tests whether the residual SSDC coefficient remains meaningful after the main governance variables are included.

In the appendix, I extend that logic to: later-adopter linear event studies for SSDC, NMPI, and TOP; static pool-family interaction models against no-pool states; carve-out interactions in addition to uniform-PDL interactions; a switcher analysis for states moving from NMPI or TOP into SSDC; outcome-construction checks using national rebate intensity and SDUD-based drug-mix controls; and an exploratory size-heterogeneity branch based on Medicaid enrollment. These analyses are explicitly secondary, designed to test whether the main result is fragile or whether a stronger positive pool story emerges elsewhere.

4.5 Inference, Cross-Checks, and Sensitivity

I use clustered standard errors at the state level throughout. To assess robustness to TWFE bias under treatment effect heterogeneity, I estimate Callaway and Sant'Anna (2021) group-time ATT parameters using not-yet-treated controls. The benchmark path reproduces closely: the ATTgt average post effect over event times 0–10 is 0.0339, compared to 0.0337 from the main pyfixest estimator, with a common-event-time correlation of 0.991 and a mean absolute event-time difference of 0.0026.

The fully controlled path does not reproduce in ATTgt because the implementation breaks on a small set of sparse change indicators and thin-support cells. Diagnostic tables isolate the failure to MCO formulary autonomy unknowns, SSDC-individual rebate-negotiation-model indicators, and vendor change flags. A reduced-control ATTgt run is estimable but does not closely reproduce the pyfixest controlled path. I therefore treat the ATTgt results as confirmation of the benchmark dynamic pattern, not as a second-software confirmation of the controlled path.

I also report Rambachan-Roth (2023) sensitivity analysis. For the uncontrolled later-adopter path, the average post-adoption estimate is 3.37 percentage points with a pointwise 95 percent confidence interval of [2.86, 3.87]. The honest confidence interval at the maximum smoothness-restricted departure ($\bar{M}$) includes zero at $M/\bar{M} = 0.06$. For the controlled path, the average post-adoption estimate is 1.28 percentage points with a pointwise interval of [0.79, 1.77], and the honest interval likewise includes zero at $M/\bar{M} = 0.06$. Both paths are fragile to even modest departures from smooth parallel trends.

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5. Results

5.1 Descriptive Trends and Timing Support

Figure 1 shows the distribution of SSDC entry years. The six early adopters (2006–2007) are already treated at the start of the post-2010 window, so the main dynamic design relies on the nine later adopters. The number of contributing treated states declines at longer event-time horizons: all 9 later adopters contribute at $k = 0$, but only 4 remain by $k = 8$ and only 2 (North Dakota and Mississippi) by $k = 10$. Long-horizon estimates should be interpreted with caution.

Figure 2 shows raw rebate trends by cohort group. SSDC cohorts tend to pull away from never-treated states over time, but the patterns are gradual

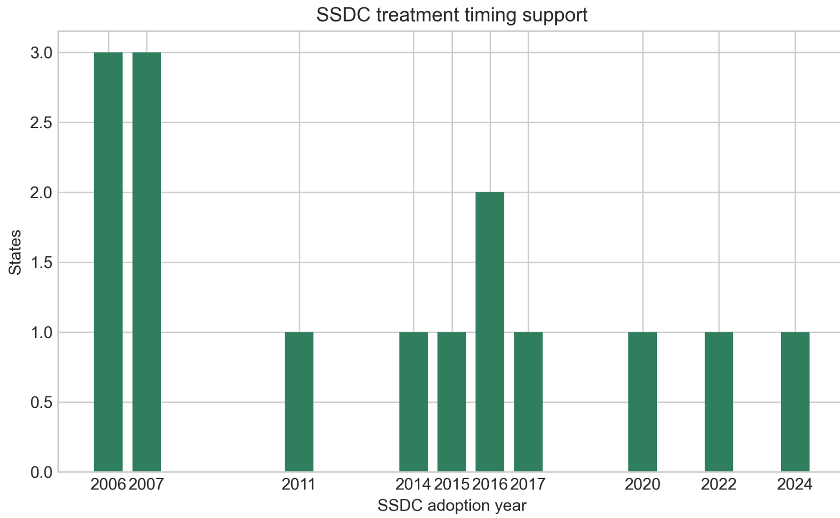

Figure 3: Figure 1. SSDC treatment timing support.

rather than sharp—consistent with an accumulating bundle of administrative changes and class-by-class renegotiation rather than a single clean rebate shock at adoption.

5.2 Later-Adopter Benchmark

In the later-adopter saturated event study, the average post-adoption effect over event times 0 through 10 is 3.37 percentage points across 660 observations and 44 states. Relative to the no-pool mean of 1.93 percent, this is a substantively large estimate—nearly a doubling of supplemental rebate capture. The uncontrolled benchmark provides the starting point against which the governance-controlled and stable-design specifications are evaluated. It should not be interpreted as a clean pool-only effect, because SSDC adoption in this sample frequently coincides with other pharmacy-benefit reforms that ordinary state and year fixed effects do not absorb.

Pre-adoption leads are generally modest: no lead from k = -10 through k = -2 exceeds 1 percentage point in absolute value, and most are statistically insignificant. Two isolated leads at k = -13 and k = -12 reach marginal significance but are far enough from the treatment window that they do not suggest a systematic pre-trend violation in the operationally relevant window.

5.3 Dynamic Pattern and Cohort Support

Before turning to the controlled specification, the dynamic pattern of the benchmark deserves examination. Effects are small and insignificant in the first two

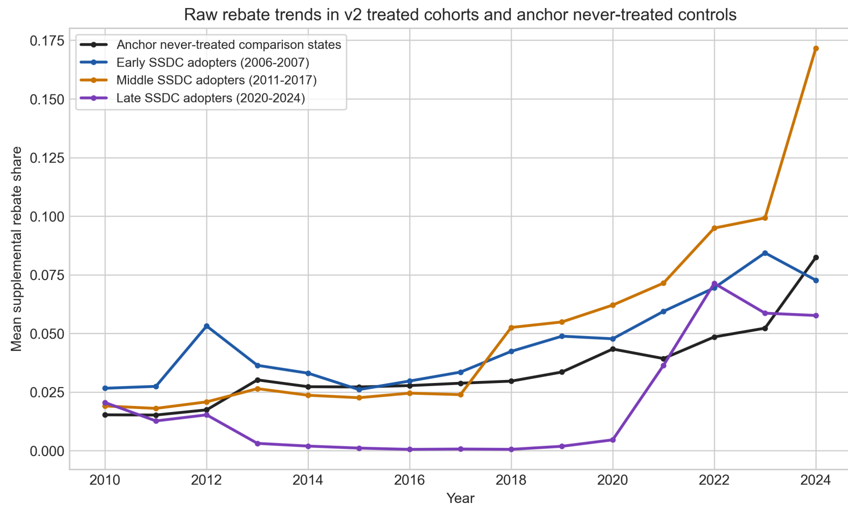

Figure 4: Figure 2. Raw rebate trends.

post-adoption years, begin to emerge at $k = 3$, and grow through longer horizons. This ramp-up is consistent with the institutional reality that supplemental rebate agreements must be renegotiated class by class: a state does not immediately renegotiate every therapeutic category the moment it enters a pool but instead cycles through its PDL at each renewal window, progressively accumulating the leverage benefits of the larger pool.

The number of contributing treated states declines at longer horizons, which matters for interpreting the dynamic path:

Event time (k)	N treated states	Contributing states
0	9	All later adopters (2011+)
5	7	Excludes PA, SD
8	4	Excludes PA, SD, KY, OH
10	2	ND, MS only

Long-horizon estimates should be interpreted with caution, as they reflect predominantly early later-adopters (smaller states with the most post-treatment years). The gradual ramp-up is informative but also non-diagnostic: the same pattern could arise from a phased governance package implemented over time, not just from progressive pool-driven renegotiation.

5.4 Governance-Controlled Main Result

The chapter's central result is that the benchmark attenuates sharply once governance controls are added. In the controlled later-adopter saturated event study, the average post-adoption effect falls from 3.37 to 1.28 percentage points across 649 observations and 44 states. Event time 0 is negative, the middle lags are only modestly positive, and the later lags become noisier.

Figure 3 shows the benchmark path alongside the controlled path. The uncontrolled path is positive on average but does not display a clean, sharp, monotone onset—the dynamic pattern is gradual and noisy.

The most important comparison in the chapter is not controlled versus zero but uncontrolled versus controlled. The difference between 3.37 and 1.28 percentage points is what the governance panel buys analytically: a large share of the

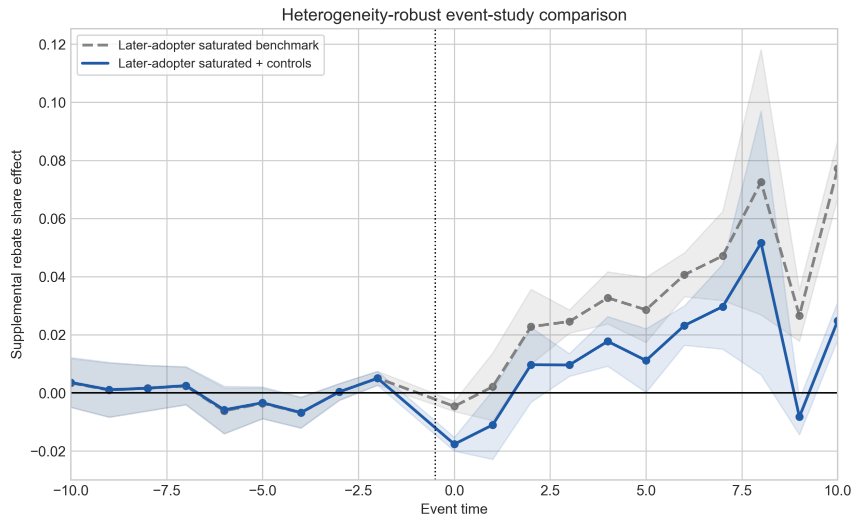

Figure 5: Figure 3. Later-adopter event-study overlay.

supplemental rebate capture; the rule that drops reform-dense windows is itself selection-on-policy-stability, which can pick up state-years that are unusually quiescent for reasons unrelated to pool membership. The useful information in the benchmark is its fragility—specifically, that the positive uncontrolled estimate does not survive the stricter design rule. Combined with the controlled estimate and the bundle model, the stable-design check supports the interpretation that the observed gains are tied to the broader governance package rather than to nominal pool entry.

The static bundle model points in the same direction.

Table 3. Main Results Summary

Specification	Obs.	States	Avg. post SSDC effect
Later-adopter saturated (2011+ adopters)	660	44	3.37 pp
Later-adopter saturated + pharmacy-benefit controls	649	44	1.28 pp

Specification	Obs.	States	Avg. post SSDC effect
Stable-design linear benchmark + controls	586	50	-2.05 pp

Table 4. Static Bundle Model

Term	Estimate	SE	p-value
SSDC adoption	1.27 pp	2.23	0.571
Uniform PDL	-1.44 pp	1.19	0.231
Pharmacy carve-out	0.48 pp	1.19	0.685
SSDC × uniform PDL	1.91 pp	1.96	0.335

Neither a robust standalone SSDC effect nor a clear complementarity story emerges from the bundle model. Adding an SSDC-by-pharmacy-carve-out interaction directly to the bundle model does not change this conclusion: the carve-out interaction is 1.27 percentage points ($p = 0.494$), and the SSDC-by-uniform-PDL term weakens further to 1.61 percentage points ($p = 0.503$).

The controlled dynamic path, the stable-design benchmark, and the bundle model all weaken the case for a large standalone SSDC effect—and they do so through different analytical lenses: the controlled path measures time-varying governance directly; the stable benchmark removes dense transition windows; the bundle model tests the full-panel static residual. The convergence across these approaches strengthens the interpretation.

5.6 ATTgt Cross-Check

The ATTgt cross-check closely confirms the uncontrolled benchmark. The ATTgt average post effect over event times 0–10 is 3.39 percentage points versus 3.37 from pyfixest, with a common-event-time correlation of 0.991 and a mean absolute event-time difference of 0.26 percentage points. The ATTgt simple aggregate estimate is 3.15 percentage points (95% CI: [1.87, 4.43]).

The fully controlled ATTgt path does not reproduce. Diagnostic tables show that the sparse vendor-change indicators, MCO formulary autonomy unknowns, and the SSDC-individual rebate-negotiation-model flag create thin-support cells that break the full ATTgt formula. A reduced-control ATTgt run with only the core uniform PDL and carve-out indicators is estimable but does not closely track the pyfixest controlled path, so the controlled result relies on one software implementation.

5.7 Rambachan-Roth Sensitivity

Both the uncontrolled and controlled average post effects are fragile to modest departures from smooth parallel trends. The Rambachan–Roth (2023) framework parameterizes the largest plausible deviation from a smooth pre-trend by a slope bound $\bar{M}$; the *honest* confidence interval at $\bar{M}$ allows the post-treatment trend to depart from the pre-treatment slope by up to $\bar{M}$ per period. For interpretability, all numbers below are reported in percentage points of supplemental rebate share.

Table R-R. Honest sensitivity bounds for the average post-adoption effect

Specification	Avg post estimate	Conventional 95% CI	Honest 95% CI at M	Breakdown M/M
Uncontrolled later-adopter	3.37 pp	[2.86, 3.87]	[-75.9, 82.6]	0.06
Controlled later-adopter	1.28 pp	[0.79, 1.77]	[-77.6, 80.1]	0.06

The maximum smoothness-restricted bound $\bar{M}$ (0.040 in raw proportion units, 4.0 pp per event-time period) is set to the largest pre-treatment slope deviation observed in the leads, so the honest interval at $\bar{M}$ permits the post-period trend to drift by as much as the noisiest pre-period drift in any direction. Under that very permissive bound the honest interval is wide because the post-treatment horizon is long. The interpretively sharper quantity is the *breakdown ratio* $M/\bar{M}$, which reports the smallest fraction of $\bar{M}$ that is sufficient to make the honest interval cover zero. Both the uncontrolled and controlled paths cross zero at $M/\bar{M}$ 0.06, meaning that even a 6 percent fraction of the largest pre-period slope deviation is enough to make the average post estimate statistically indistinguishable from zero. That is a stringent fragility result, not because the underlying point estimate is small but because the post-treatment horizon is long enough that small drift accumulates.

5.8 Negative Selection Evidence

Although not the main design, the descriptive comparison between SSDC and NMPI/TOP state-years is informative about selection. In raw data, future SSDC adopters had systematically *lower* supplemental rebate percentages than NMPI/TOP states in the years before joining SSDC. The pool-family interaction models in Section 5.9 show that the standalone SSDC coefficient against no-pool states is only 0.91 percentage points (p = 0.658), while the standalone NMPI coefficient is -3.74 percentage points (p = 0.008). These patterns suggest negative selection into SSDC: states joined because their existing rebate capture

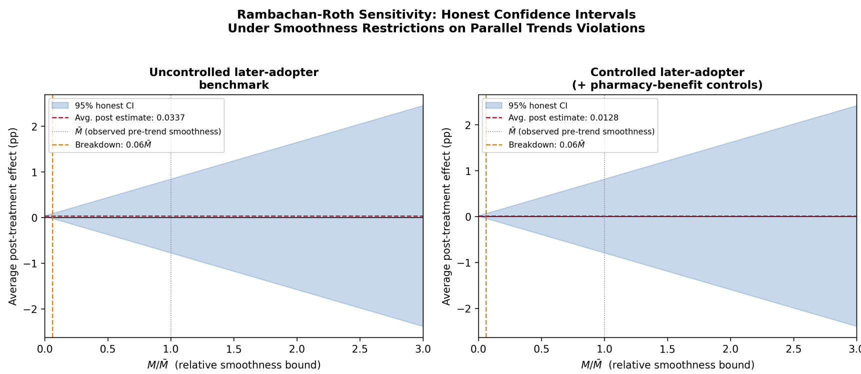

Figure 6: Figure 4. Rambachan-Roth sensitivity bounds.

was inadequate, not because they were already high performers. If this negative selection extends to the main later-adopter specifications, it biases estimates against finding positive effects, which makes the residual positive benchmark (before governance controls) more noteworthy—and makes the attenuation after governance controls all the more informative about what was really driving the observed gains.

5.9 Broader Pool-Family Extensions

A natural question is whether the SSDC result weakens only because the chapter is looking in the wrong place. Controlled linear later-adopter event studies yield average post estimates of 2.24 percentage points for SSDC, -0.95 percentage points for NMPI, and -0.53 percentage points for TOP. Saturated later-adopter extensions are similarly flat or negative outside SSDC: uncontrolled average post effects are -1.58 percentage points for NMPI (2 treated states) and -1.33 percentage points for TOP (3 treated states). The controlled TOP saturated estimator fails entirely due to sparse support.

Table 5. Pool-Family Later-Adopter Extensions

Pool	Spec.	Obs.	States	Treated	Avg. post
SSDC	Linear controlled	619	43	9	2.24 pp
NMPI	Linear controlled	504	35	2	-0.95 pp
TOP	Linear controlled	589	41	3	-0.53 pp
SSDC	Saturated uncontrolled	660	44	9	3.37 pp
SSDC	Saturated controlled	649	44	9	1.28 pp
NMPI	Saturated uncontrolled	525	35	2	-1.58 pp
TOP	Saturated uncontrolled	630	42	3	-1.33 pp

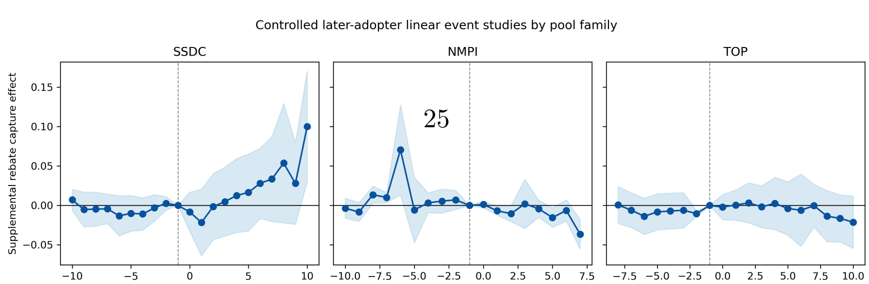

Figure 7: Figure 5. Pool-family event studies.

PDL = 3.48 pp (p = 0.054), NMPI × uniform PDL = 3.04 pp (p = 0.016), TOP × uniform PDL = 2.84 pp (p = 0.054), while the standalone pool coefficients are small or negative. But once carve-out interactions are added, the uniform-PDL interactions become imprecise (SSDC × uniform PDL falls to 3.03 pp, p = 0.301; NMPI to 2.86, p = 0.176; TOP to 2.39, p = 0.425), and the carve-out interactions themselves are all near zero and insignificant.

5.10 Placebo Test

A shifted-date placebo test assigns pseudo-treatment dates three years earlier than actual adoption. Under the null of no effect, the pseudo post-period should show no effects. The placebo results are not fully reassuring: several pre-placebo-date coefficients are positive and significant (the k = -9 pseudo-lead is 4.86 pp, p < 0.001; k = -8 is 3.59 pp, p = 0.001), showing positive pseudo-effects before the placebo date. This further weakens the old benchmark onset story, though the shifted-date placebo is a legacy diagnostic tied to the original TWFE benchmark rather than the new saturated later-adopter estimator. The positive pseudo-pre-period coefficients are consistent with accumulating governance changes rather than a clean treatment onset.

5.11 Federal-Rebate Headroom and Drug-Mix Checks

Two critiques target the supplemental rebate outcome itself. The headroom critique holds that states with high federal rebate shares may have less post-federal spending from which to collect supplemental rebates. The drug-mix critique holds that higher supplemental rebate percentages could reflect basket composition rather than stronger negotiation.

Table 6. Outcome-Construction Robustness

Bundle specification	Outcome	Obs.	SSDC coef.	p-value
Baseline	Supp. rebate share	709	1.27 pp	0.571
+ national rebate share	Supp. rebate share	709	1.47 pp	0.496
+ national rebate + SDUD mix	Supp. rebate share	709	1.54 pp	0.476
Post-federal outcome	Supp./post-federal	706	4.60 pp	0.544

Conditioning on national rebate intensity does not rescue a positive standalone SSDC effect. In fact, national and supplemental rebate shares are positively correlated in the panel rather than mechanically offsetting one another. Redefining the outcome as supplemental rebates divided by post-federal spending also does not change the interpretation.

SDUD-based drug-mix controls tell a similarly bounded story. SSDC state-years do differ from no-pool states descriptively—the SSDC single-source spend share is 45.5 percent versus 24.5 percent for no-pool states, and average spend per prescription is $91.46 versus $79.72. But adding those mix measures to the bundle model leaves the SSDC coefficient small and imprecise. The main null is not well explained by cross-state differences in drug usage mix.

The later-adopter linear benchmark tells the same story: adding national rebate share changes the average post effect from 2.24 to 2.01 percentage points; adding SDUD mix controls returns it to 2.24 percentage points.

5.12 Switchers and Size Heterogeneity

Five states switch from NMPI or TOP into SSDC: Delaware, Kentucky, Pennsylvania, Vermont, and West Virginia. Relative to 22 states that ever participate in NMPI or TOP but never join SSDC, the post-switch SSDC estimate is 3.15 percentage points in a parsimonious fixed-effects model ($p = 0.043$) and 3.02 percentage points with fuller governance controls ($p = 0.164$). The switcher analysis is suggestive but not stable: the estimate stays positive after adding governance controls but is no longer statistically precise, and the switcher sample is small enough that individual state outliers can meaningfully shift the result.

5.13 Pool-Size Heterogeneity (Appendix-Style Branch)

This subsection reports an exploratory branch that the main contribution does not turn on. It is retained in the main text for transparency, but readers should treat the SSDC relative-size result as a fragile descriptive contrast — driven primarily by Ohio (an entrant much larger than the incumbent pool leader) and disappearing under the most plausible sample restrictions. A natural extension asks whether smaller Medicaid programs benefit disproportionately from pool membership. The theoretical logic is straightforward: a small state joining a pool with large incumbents may gain access to bargaining leverage it could not generate on its own, while a large state may already possess sufficient covered lives to negotiate effectively without a pool. If pooling works primarily by lending scale to small programs, the effect should be larger for smaller entrants.

I test this hypothesis using a separate size-heterogeneity branch that constructs pre-entry Medicaid enrollment measures for all pool entrants and estimates whether entrant size moderates the pool-membership effect. The enrollment panel covers 2002–2024 and bridges two sources: CMS-64 enrollment data for 2002–2013 and official MBES (Medicaid Beneficiary Extract Summary) monthly data collapsed to annual averages for 2014–2024. A validation check on the 2014–2020 overlap period confirms the two sources are closely aligned (mean ratio = 1.004).

For each pool entry, I measure entrant size in two ways. The *absolute-size* specification uses log pre-entry enrollment (centered), which captures whether larger or smaller programs in general benefit more. The *relative-size* specification uses entrant smallness, defined as one minus the ratio of the entrant's enrollment to the largest incumbent member's enrollment at the time of entry. This measure is only defined for non-first-wave entrants—states that joined a pool after at least one incumbent was already present—and it directly tests the leverage-borrowing hypothesis.

Table 7. Pool-Size Heterogeneity Results

Pool	Specification	Interaction	SE	p-value	Obs.	States	Treated
SSDC	Relative size (smallness)	1.70 pp	0.61	0.008	1,042	47	12
SSDC	Absolute size (log enroll.)	-0.31 pp	0.65	0.634	1,111	50	15
NMPI	Relative size (smallness)	0.06 pp	0.58	0.922	1,042	47	14
NMPI	Absolute size (log enroll.)	-0.54 pp	0.63	0.392	1,111	50	17
TOP	Relative size (smallness)	1.02 pp	4.10	0.804	996	45	6
TOP	Absolute size (log enroll.)	0.48 pp	0.91	0.598	1,111	50	11

SSDC shows the only statistically significant result: the relative-size interaction is 1.70 percentage points ($p = 0.008$), suggesting that smaller SSDC entrants gain more supplemental rebate capture than larger ones, conditional on entering after an incumbent is already established. The absolute-size specification for SSDC is null (-0.31 pp, $p = 0.634$), indicating that the pattern is specific to entrant size *relative to the largest incumbent*, not to program size in general. NMPI and TOP show no comparable pattern in either specification.

However, the SSDC relative-size result is fragile under closer examination. Three robustness checks reveal the sensitivity:

Table 8. SSDC Relative-Size Robustness

Variant	Treated	Interaction	p-value
Baseline	12	1.70 pp	0.008
Exclude Kentucky (partial-year MBES)	11	1.77 pp	0.007
Exclude entrants larger than incumbent	8	-2.26 pp	0.729

Dropping Kentucky (which entered in 2024 with only nine months of pre-entry MBES coverage) slightly strengthens the signal. But restricting the sample to genuinely small entrants—those with entry smallness at or above zero, meaning they were actually smaller than the largest incumbent—eliminates the result entirely (-2.26 pp, $p = 0.729$, 8 treated states).

A leave-one-out analysis identifies the source of the fragility. Excluding Ohio drops the interaction from 1.70 to 0.54 percentage points ($p = 0.787$). No other state exclusion moves the coefficient by more than 0.5 percentage points. Ohio's outsized influence is not surprising: it entered SSDC in 2017 with 3.1 million Medicaid enrollees, nearly three times larger than Oregon (the largest incumbent at that time with 1.05 million), giving it an entry smallness of -1.94. Ohio is an entrant *larger* than the incumbent pool leader, so its inclusion creates a wide contrast between a very large entrant and genuinely small ones. When that contrast is removed, the gradient flattens.

The SSDC relative-size result is therefore better interpreted as a contrast between Ohio (a very large late entrant with modest post-entry rebate gains) and a handful of smaller late entrants (some of which show stronger gains) rather than as evidence of a clean, generalizable small-state advantage. The absence of any comparable pattern in NMPI or TOP further limits the generalizability of the finding. I report these results transparently but do not treat them as a headline finding.

5.14 Any-Pool Specification and Triple Interactions

The main analysis distinguishes SSDC, NMPI, and TOP as separate pool families. A natural robustness check collapses that distinction and asks whether membership in *any* interstate pool increases supplemental rebate capture, regardless of which pool. This any-pool indicator also permits cleaner tests of pool-by-governance interactions without splitting the already-small treated samples across three pool families.

Table 9. Any-Pool Models

Model	Term	Estimate	SE	p-value
(1) Any-pool standalone	Any pool	0.06 pp	0.98	0.951
(2) + governance controls	Any pool	0.07 pp	1.02	0.946
	Uniform PDL	-0.60 pp	1.24	0.630
	Pharmacy carve-out	0.98 pp	1.08	0.370
(3) + any-pool × uniform PDL	Any pool	-1.19 pp	0.73	0.110
	Uniform PDL	-2.26 pp	1.81	0.222
	Any pool × uniform PDL	2.31 pp	1.28	0.078
(4) + carve-out interaction	Any pool	-1.26 pp	0.79	0.116
	Any pool × uniform PDL	0.69 pp	1.78	0.700
	Any pool × carve-out	2.56 pp	2.14	0.241
(5) Triple interaction	Any pool	-0.42 pp	0.69	0.550
	Any pool × uniform PDL	-1.69 pp	1.32	0.209
	Any pool × carve-out	-4.01 pp	1.10	0.001
	Any pool × PDL × carve-out	**8.91 pp**	**2.99**	**0.005**

Notes: All models include state and year FE with state-clustered SEs. N = 709 (complete governance sample, 49 states). Model (5) drops the two-way uniform PDL × carve-out interaction due to collinearity.

The any-pool standalone coefficient is effectively zero (0.06 pp, $p = 0.951$), confirming the main paper's conclusion: pooling by itself does not raise supplemental rebate capture. Adding governance controls (Model 2) does not change this. The any-pool-by-uniform-PDL interaction (Model 3) is suggestive but imprecise (2.31 pp, $p = 0.078$), and it becomes statistically weak once the carve-out interaction is also included (Model 4: 0.69 pp, $p = 0.700$).

The triple-interaction model (Model 5) produces the most suggestive descriptive result. The three-way interaction—any pool × uniform PDL × pharmacy carve-out—is 8.91 percentage points ($p = 0.005$). This coefficient captures states that simultaneously (a) participate in an interstate pool, (b) maintain a uniform PDL, and (c) have carved the pharmacy benefit out of managed care. These are states with the most centralized, integrated pharmacy-governance architecture.

The finding is consistent with the chapter's bundled-governance interpretation: the strongest supplemental rebate performance emerges when all three institutional features are present together, not when any one feature is adopted in isolation. The standalone any-pool coefficient and the two-way interactions are all small and insignificant, suggesting that pooling, uniform PDLs, and carveouts individually accomplish little. The combination is consistent with greater rebate capture, but this static interaction does not by itself identify causal complementarity among the three levers.

This result should be interpreted cautiously for two reasons. First, the triple interaction is identified from a specific subset of state-years where all three conditions are simultaneously active, and the share of identifying variation contributed by any one state matters for credibility. Second, the model drops the two-way uniform PDL × carve-out interaction due to collinearity with the other terms, which means the triple interaction absorbs variation that might otherwise be attributed to a governance-only (non-pool) channel.

To address the first concern directly, Table 9a reports the support cells for the all-three-features (any-pool × uniform PDL × pharmacy carve-out = 1) combination. Identification draws on 181 state-years across 19 states between 2006 and 2024, including substantial contributions from Idaho, Maine, Montana, Rhode Island, Vermont, Wisconsin, and Wyoming, plus shorter contributions from later switchers (Mississippi, Ohio, Oklahoma, West Virginia, Pennsylvania-via-NMPI). The cell is not concentrated in one or two states.

Table 9a. Triple-Interaction Support — States Identifying the All-Three Cell

State	State-years in cell	Year range	Mean supp. rebate share
Arkansas	4	2021–2024	4.87%
Connecticut	13	2012–2024	4.60%
Idaho	15	2010–2024	4.72%
Iowa	10	2006–2015	5.59%
Maine	15	2010–2024	5.74%
Mississippi	1	2024–2024	19.73%
Montana	15	2010–2024	3.26%
New York	2	2023–2024	7.29%
North Carolina	11	2010–2020	4.42%
North Dakota	5	2020–2024	9.35%
Ohio	3	2022–2024	8.74%
Oklahoma	8	2016–2023	8.78%
Oregon	3	2010–2012	0.15%
Rhode Island	15	2010–2024	0.27%
Tennessee	8	2013–2020	6.02%
Vermont	15	2010–2024	7.15%
West Virginia	8	2017–2024	7.04%
Wisconsin	15	2010–2024	4.42%
Wyoming	15	2010–2024	6.44%

Cell-level descriptives reinforce the bundled-governance reading: the all-three cell averages 5.23% supplemental rebate share, the pool-with-PDL-only cell averages 3.95%, the pool-with-carve-out-only cell is not informatively populated (2 state-years in 1 state), and the no-pool-no-PDL-no-carve-out cell averages

1.55%. The pool-only-with-no-other- governance cell averages 2.53%, close to the no-pool baseline. The adjacent no-pool-with-PDL-and-carve-out cell averages 3.52% across 57 state-years in 8 states, so the clean descriptive gap between the all-three cell and the same-governance no-pool cell is closer to 1.7 percentage points; the residualized adjacent-cell contrast is 2.79 pp. That comparison is smaller than the full triple-interaction coefficient and is an important reminder that part of the large three-way coefficient reflects sparse support and the dropped two-way uniform-PDL-by-carve-out term.

A leave-one-state-out analysis confirms that the triple interaction is not driven by one or two unusual states among the estimable leave-one-out specifications. For 48 of the 49 leave-one-state-out runs, the triple-interaction estimate stays positive and remains statistically significant ($p < 0.05$). The South Dakota leave-one-out specification fails because the remaining design matrix loses support for the required interaction cell, which is itself a useful support warning. The largest movers among the estimable specifications are:

Dropped state	Estimate	p-value	Δ vs full sample
(Full sample)	8.91 pp	0.005	–
Drop Oklahoma	6.06 pp	0.004	-2.85 pp
Drop Arkansas	10.99 pp	0.000	+2.08 pp
Drop Virginia	10.55 pp	0.001	+1.64 pp
Drop Minnesota	7.78 pp	0.008	-1.13 pp
Drop Connecticut	9.32 pp	0.006	+0.41 pp

No other state moves the estimate by more than 1 percentage point. Oklahoma is the most influential single state, but even excluding it, the estimate remains 6.06 pp at $p = 0.004$. The result therefore does not collapse to a "one or two unusual states" story among estimable leave-one-out specifications, while the South Dakota failure reinforces the need to interpret the triple-interaction model as a support-limited descriptive exercise.

With these caveats and diagnostics in view, the triple-interaction finding is best read as a *support-limited descriptive concentration result*: among the relatively few state-years in which pool participation, a uniform PDL, and a pharmacy carve-out are all simultaneously active, supplemental rebate capture is meaningfully higher than in adjacent governance combinations. It is not the chapter's headline causal finding. The headline is the *attenuation pattern* across the three estimands defined in §4.1 — uncontrolled benchmark (3.37 pp) → governance-controlled (1.28 pp) → stable-design benchmark (–2.05 pp) — and the near-zero standalone any-pool coefficient (0.06 pp). The triple interaction sharpens the descriptive geography of where any positive bundle performance concentrates; it does not establish that adding pool membership to a uniform-PDL/carve-out regime causes the additional rebate gains.

6. Discussion

6.1 Interpretation and Policy Implications

The chapter's central finding is that the positive SSDC benchmark attenuates materially once pharmacy-governance changes are measured directly. The evidence points away from a clean pool-only story and toward a broader bundled-governance interpretation. States often discuss interstate pools as if pool membership were a clean, portable lever for higher supplemental rebate capture. The evidence here suggests that view is too simple.

If states realize stronger supplemental rebate capture around SSDC adoption, those gains appear more plausibly tied to a package that may include centralized formulary control, changes in managed-care pharmacy organization, and altered rebate administration. The triple-interaction analysis reinforces this interpretation: the only statistically significant positive coefficient in the any-pool specifications is the three-way interaction between pool membership, a uniform PDL, and a pharmacy carve-out (8.91 pp, $p = 0.005$), identified from 181 state-years across 19 states and robust across the estimable leave-one-state-out specifications. The adjacent no-pool but uniform-PDL-and-carve-out cell has meaningfully lower average rebate capture, but the clean adjacent-cell gap is smaller than the full triple-interaction coefficient. Pooling, uniform PDLs, and carve-outs each accomplish little in isolation; the combination is where rebate gains appear most concentrated. A state considering whether to join a pool should first ask whether it has the administrative architecture needed to translate pooled covered lives into real negotiating leverage. If its PDL remains fragmented across plans, if rebate authority is diffuse, or if major governance reforms have not been made, simply joining a pool may not reproduce the observed gains.

Rebate capture is not the same as net savings. A formulary strategy that increases supplemental rebate dollars by steering utilization toward high-rebate branded drugs may also raise gross drug spending or reduce generic substitution. Munshi et al. (2018) document precisely this pattern in Florida, where a state-mandated PDL reduced overall drug use by 6 percent but increased plan costs by 27 percent because the PDL channeled utilization toward preferred branded drugs with high acquisition costs but high rebate yields. The outcome studied here—supplemental rebate share—captures one fiscally meaningful component of Medicaid pharmacy finance, but it is not the full welfare or budgetary effect of pool participation. Throughout the chapter I therefore use the language of "supplemental rebate capture" rather than "savings."

While the analysis cannot definitively identify why SSDC descriptively outperforms NMPI and TOP, the institutional evidence points to several structural features:

Individual contracting with pooled leverage. SSDC's signature feature is that each member state contracts individually with manufacturers while leveraging the consortium's combined covered lives. This preserves state autonomy over the negotiation while amplifying bargaining position. In contrast, NMPI and TOP negotiate collective agreements that states opt into drug by drug. The individual contracting model may be more effective because it aligns the state's incentives directly with aggressive negotiation: the state bears the full cost of weak negotiation and captures the full benefit of strong negotiation, avoiding the free-rider problem inherent in collective models.

PDL autonomy. SSDC member states retain complete control over their PDLs, allowing formulary placement decisions tailored to their population's clinical needs and class-specific competitive dynamics. This flexibility may enable SSDC states to exploit class-specific leverage opportunities that a uniform pool-wide PDL cannot capture.

Knowledge sharing. SSDC provides substantial market intelligence, benchmarking data, and negotiating support—institutional knowledge transfer particularly valuable for smaller states with limited pharmaceutical negotiating capacity. This is consistent with Dubois, Lefouili, and Straub's (2021) finding that centralized procurement lowered drug prices, while gains were smaller where supplier concentration limited buyer leverage.

These structural hypotheses are consistent with Ho and Lee's (2024) emphasis on the ability to steer utilization rather than simply accumulate covered lives, and with Parmaksiz et al.'s (2022) emphasis on buyer capacity, organizational capacity, supplier incentives, and operational design. The chapter's results fit that literature better than a simple "bigger pool equals bigger rebates" story.

6.2 Why a Null-Leaning Result Is a Useful Contribution

Interstate pools have often been discussed with limited causal evidence and strong implied savings claims. Demonstrating that those claims do not survive direct measurement of the most plausible bundled reforms is valuable information for states deciding how to allocate scarce administrative attention.

The chapter also contributes methodologically by illustrating how an apparently promising Medicaid policy result can change once the design engages the actual administrative bundle. In many state-policy settings, the policy indicator of interest is really a summary label for a cluster of governance changes. The lesson from this case generalizes: a credible causal design often requires measuring the surrounding administrative architecture rather than assuming fixed effects will absorb it.

6.3 Illustrative Supplemental-Rebate Projections and Their Limits

These projections are mechanical and should not be interpreted as expected incremental supplemental-rebate dollars from pool adoption alone. The chapter's main result is that the standalone pool effect is weak; extrapolations based on pool membership alone will overstate the effect of nominal pool entry. They are reported here only to bound the order of magnitude that policy discussions sometimes invoke.

Fifteen SSDC states currently account for approximately \$20 billion of the \$80 billion in gross Medicaid drug spending. If the full uncontrolled benchmark (3.37 pp) were taken at face value and applied to the remaining \$60 billion, the implied additional supplemental-rebate dollars would be roughly \$2.0 billion annually. Using the controlled estimate (1.28 pp) instead yields \$0.77 billion. With a 50 percent attenuation for general-equilibrium responses—manufacturer pricing adjustments, diminishing marginal leverage from universal adoption—these projections narrow to \$0.38–1.0 billion annually. Because the standalone any-pool coefficient is essentially zero, the lower bound (and arguably zero) is the more policy-defensible projection.

Two more reasons argue for caution. First, "rebate capture" is not "net savings": as discussed in §6.1, a formulary strategy that increases rebates by favoring high-rebate branded drugs may also raise gross spending. Second, extrapolating from 15 SSDC adopters to 35 non-SSDC states assumes the observed SSDC advantage is not largely capturing concurrent reforms—an assumption the chapter's main analysis directly questions.

6.4 Relation to Prior Literature

This chapter fills a gap at the intersection of several literatures. The Medicaid pharmacy benefit literature has examined managed care's effects on drug spending (Dranove, Ody, and Starc, 2021), pharmacy carve-outs (Bendicksen and Kesselheim, 2022), and PDL mandates' effects on utilization and costs (Munshi et al., 2018; Virabhak and Shinogle, 2005), but has not examined the role of interstate purchasing pools in supplemental rebate capture once pharmacy-governance structure is measured directly. The pharmaceutical procurement literature has found that pooled procurement can reduce prices in international settings (Dubois, Lefouili, and Straub, 2021) and that governance design is a key determinant of success (Parmaksiz et al., 2022), but has not tested those predictions in the U.S. Medicaid rebate context.

The result is more nuanced than a simple confirmation of the pooled-procurement literature. It suggests that pooled bargaining may matter but is difficult to separate from the governance bundle through which states actually operationalize pharmacy control.

6.5 Limitations

Several limitations remain. First, some historical pharmacy-benefit spells are medium-certainty backcasts rather than archive-verified year-by-year coding. North Dakota, Ohio, and Kentucky have the most important remaining archival-validation gaps. Transition dates are annual rather than monthly. Vendor identity is sparser than the core governance variables.

Governance controls are not a clean fix for confounding. As discussed in §4.1, the time-varying pharmacy-benefit controls play either a confounding role or a mediating role—and likely both, depending on the variable. Where uniform PDL adoption or carve-out reform is genuinely pre-existing or independently concurrent with SSDC entry, conditioning on it helps recover a direct pool effect. Where these reforms are part of how a state operationally implements pool participation, conditioning on them blocks part of the treatment pathway and yields a controlled-direct-effect estimand rather than the total policy effect. The controlled estimates therefore identify a residual pool association conditional on observed governance architecture, not the total effect of adopting a pool as states implement it in practice. A reviewer concerned about "bad controls" is correct on the technical point; the chapter's response is to report both the uncontrolled and controlled estimates and to interpret their *difference* as the bundle effect, rather than to claim the controlled estimate is the single right answer.

The governance-controlled estimate is identified almost entirely on the post-2010 window. As Table 2d shows, the complete-controls sample is 98.9% post-2010 state-years; pre-2010 state-years drop out for lack of governance coding. The controlled estimate is therefore best read as the residual SSDC association in the modern post-MDRP-expansion landscape, not over the full 2002–2024 panel.

Second, the controlled later-adopter path relies on one software implementation because ATTgt fails when the full time-varying control set is added. The ATTgt diagnostic tables isolate the failure to sparse change indicators and thin-support cells, and a reduced-control ATTgt run does not closely reproduce the pyfixest controlled path. This single-software dependence is a real limitation for a submission-facing result.

Third, NMPI and TOP have only two and three later adopters with complete control support, so those family-level comparisons remain exploratory rather than definitive. The broader pool-family extensions should be understood as testing whether the main null is fragile, not as precisely estimating family-specific effects.

Fourth, the SDUD category crosswalk matches spending much better in later years than in the early 2000s, so the drug-mix checks are most informative in the modern window.

Fifth, there is an unavoidable limit to what this state-year design can reveal about mechanism. The governance controls indicate whether major administrative features changed near pool entry, but they do not observe class-level rebate terms, manufacturer-specific bargaining responses, or plan-level enforcement of preferred placement. A future design with class-level rebate or utilization management data could do more to distinguish whether residual variation comes from bargaining skill, therapeutic mix, contract architecture, or measurement noise.

Finally, the Rambachan-Roth sensitivity analysis shows that both the uncontrolled and controlled average post effects include zero under even modest departures from smooth parallel trends, further cautioning against overselling any residual positive SSDC coefficient.

6.6 Future Research

Several next steps follow naturally. First, deeper archival validation of medium-certainty policy spells, especially for historically important treated and comparison states, would strengthen the governance panel's credibility for journal submission. Second, richer state-year data on managed-care pharmacy operations and rebate administration—ideally standardized and publicly reported by CMS or MACPAC—would enable more precise identification. Third, drug-class-level work that could directly connect formulary structure, therapeutic competition, and supplemental rebate outcomes would address the mechanism question that state-year data cannot reach. Fourth, as later adopters (particularly Pennsylvania) accumulate more post-treatment years, the broader pool-family dynamic extensions will become more informative. Finally, improved CMS-64 reporting—standardized line-item definitions, data validation checks, and documentation of reporting changes across fiscal years—would reduce the outcome measurement noise that currently limits precision.

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7. Conclusion

This chapter provides a more credible test of interstate Medicaid drug purchasing pools than a simple pool-membership event study. Once pharmacy-benefit structure is measured directly and treatment timing is handled carefully, the evidence does not support a large standalone SSDC effect on supplemental rebate capture. The uncontrolled later-adopter benchmark (3.37 percentage points) attenuates to 1.28 percentage points with governance controls, while the stable-design benchmark is negative (-2.05 pp), the bundle-model standalone SSDC coefficient is small and imprecise (1.27 pp, $p = 0.571$), and Rambachan-Roth honest intervals include zero under minimal departures from smooth parallel trends.

The appendix extensions do not reveal a stronger alternative story. NMPI and TOP do not show cleaner positive dynamics, carve-out interactions do not rescue a pool effect, switchers provide only suggestive evidence, and federal-rebate headroom and drug-mix critiques do not explain the null. A pool-size heterogeneity branch shows only a fragile SSDC relative-size pattern driven by Ohio that does not replicate across pools or survive the most plausible sample restrictions; I report it transparently but do not treat it as a headline finding. The any-pool standalone coefficient is effectively zero (0.06 pp, $p = 0.951$), confirming the null generalizes beyond SSDC. As a support-limited descriptive concentration result, the triple-interaction finding — pool $\times$ uniform PDL $\times$ pharmacy carve-out, identified from 181 state-years across 19 states — is consistent with the bundled-governance reading: supplemental rebate capture concentrates where all three governance features are simultaneously active, not in any single component on its own. But the triple interaction is not the paper's headline; the headline is the attenuation across the three estimands.

For policy, the immediate implication is caution. States should not assume that joining SSDC, NMPI, or TOP by itself will reproduce the strongest observed rebate outcomes. Pool membership should be evaluated as one component of a broader administrative package—including uniform PDL governance, pharmacy carve-out structure, and rebate-negotiation architecture—rather than as a standalone lever for higher supplemental rebate capture (and, separately, should not be expected to deliver net drug-spending savings without analysis of net spending). Federal policymakers should improve transparency around pool participation and supplemental rebate outcomes to enable ongoing evaluation and evidence-based purchasing decisions.

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Appendix

Appendix A. Pharmacy-Benefit Variables

Variable	Description	Why it matters
`uniform_pdl`	State has a uniform preferred drug list spanning relevant delivery systems	Can increase formulary leverage and rebate bargaining power
`pharmacy_carve_out`	Pharmacy benefit carved out of managed care	Changes who controls the rebate and utilization-management structure
`mco_formulary_autonomy`	Degree of managed-care formulary independence	Captures whether managed care plans dilute state leverage
`pdl_vendor`	Preferred drug list support vendor or administrator	Administrative capacity and contracting style may change with vendors
`pbm_vendor`	Pharmacy benefit manager or analogous contractor	May shift negotiation execution and reporting
`rebate_negotiation_model`	State direct, SSDC individual, collective pool, PBM-assisted, hybrid, or unknown	Distinguishes pool membership from how bargaining is actually done
`major_reform_flag`	Major administrative or policy transition in a spell	Used to define stable-design windows

Appendix B. Current Results Table

Specification	Observations	States	Average post-adoption SSDC effect
Later-adopter saturated event study (2011+ adopters)	660	44	0.0337
Later-adopter saturated event study + pharmacy-benefit controls	649	44	0.0128
Stable-design linear benchmark + pharmacy-benefit controls	586	50	-0.0205

Bundle model term	Estimate	Standard error	p-value
`post_ssdc`	0.0127	0.0223	0.571
`uniform_pdl`	-0.0144	0.0119	0.231
`pharmacy_carve_out`	0.0048	0.0119	0.685
`ssdc_x_uniform_pdl`	0.0191	0.0196	0.335

Appendix C. Why State and Year Fixed Effects Are Not Enough

State fixed effects absorb time-invariant differences across states, and year fixed effects absorb shocks common to all states in a given year. They do not absorb within-state, time-varying policy changes such as adopting a uniform preferred drug list, changing carve-out status, altering formulary autonomy, or shifting rebate-negotiation models. If those reforms occur near SSDC entry and also affect supplemental rebate capture, their effects can still load onto the SSDC coefficient in a model that includes only state and year fixed effects. That is why the added pharmacy-benefit panel matters substantively rather than cosmetically.

Appendix D. Stable-Design Rule

The current implementation defines `stable_design_pm2 = 1` when a state-year is not within a plus/minus two-year window of a major benefit transition. Major transitions include:

- explicit `major_reform_flag = 1`
- a change in uniform PDL status
- a carve-out or carve-in switch
- a change in PDL vendor
- a change in PBM vendor
- a change in rebate negotiation model

Appendix E. Current Coverage Snapshot

Metric	Value
Coded state-years in pharmacy-benefit panel	747
States covered	50
Never-treated comparison states with complete controls	35
Share of merged analytic file with complete key controls	0.6496

Appendix F. Remaining Limitations

- Several historical policy spells remain medium-certainty backcasts rather than archive-verified year-by-year coding.
- Transition dates are annual rather than monthly.
- Kansas portal-linked source paths should be validated in a browser before final external submission.
- Vendor identity remains sparse and should not be overinterpreted.
- The benchmark later-adopter path has an ATTgt second-software cross-check that gives an average post effect of `0.0339` over event times `0-10` versus `0.0337` in `pyfixest`.
- ATTgt currently fails once the full time-varying pharmacy-benefit control set is added, so the controlled later-adopter path still relies on `pyfixest`.
- The controlled later-adopter dynamic sample is materially broader than earlier passes at 649 observations across 44 states, though it still relies on one software implementation.
- NMPI and TOP have only two and three later adopters with complete control support in the broader dynamic extensions, so those family-level comparisons are exploratory.
- SDUD category-share controls use a 2025 rebate-drug source file whose matched spend share rises from 26.6 percent in 2002 to 99.5 percent in 2024, so those category-based mix checks are most informative in the later part of the panel.

Appendix G. Figure Package

Corrected treatment timing shows six early SSDC adopters in 2006-2007, so the main dynamic estimator uses only the nine later adopters from 2011-2024 plus never-treated controls. The shifted-date placebo figure below is a background diagnostic rather than the paper's main event-study estimator.

The mechanism atlas below unpacks the governance channels referenced in the main text.

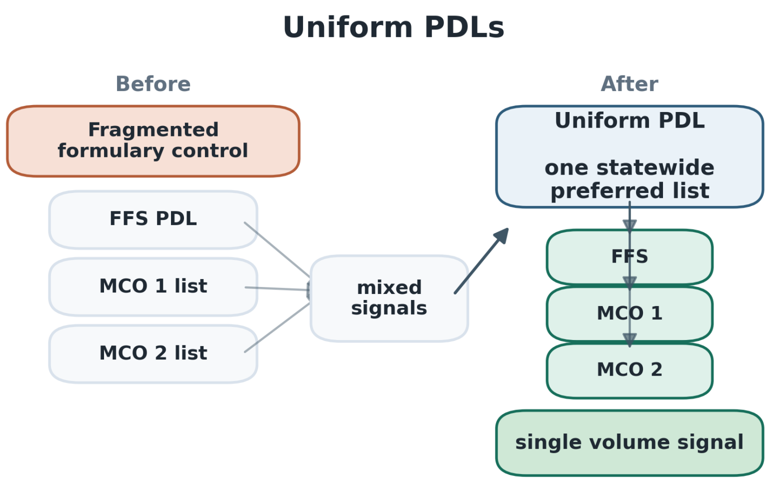

Figure 8: Uniform PDLs

Appendix H. Later-Adopter Dynamic Sample

The saturated event-study estimator uses the nine SSDC cohorts first treated in 2011 or later:

- North Dakota (2011)
- Mississippi (2014)
- West Virginia (2015)
- Delaware (2016)
- Oklahoma (2016)
- Ohio (2017)
- Pennsylvania (2020)
- South Dakota (2022)
- Kentucky (2024)

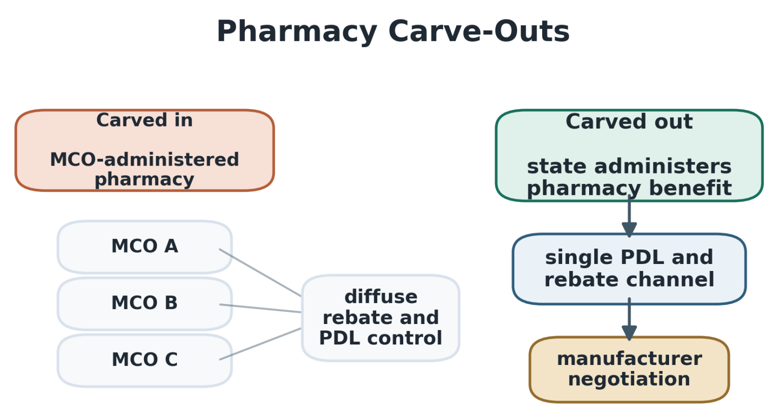

Figure 9: Pharmacy carve-outs

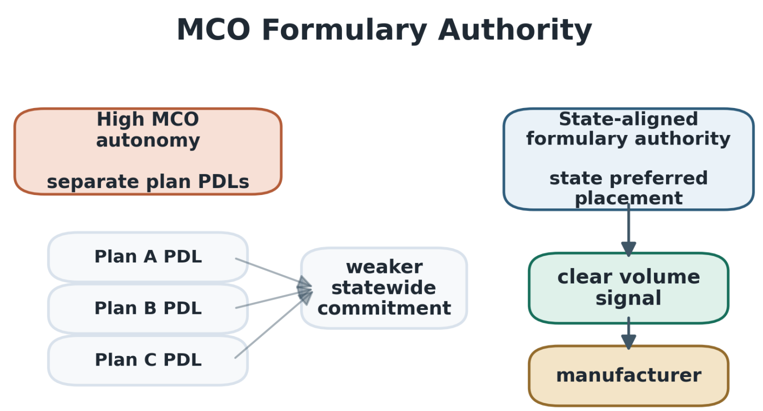

Figure 10: MCO formulary authority

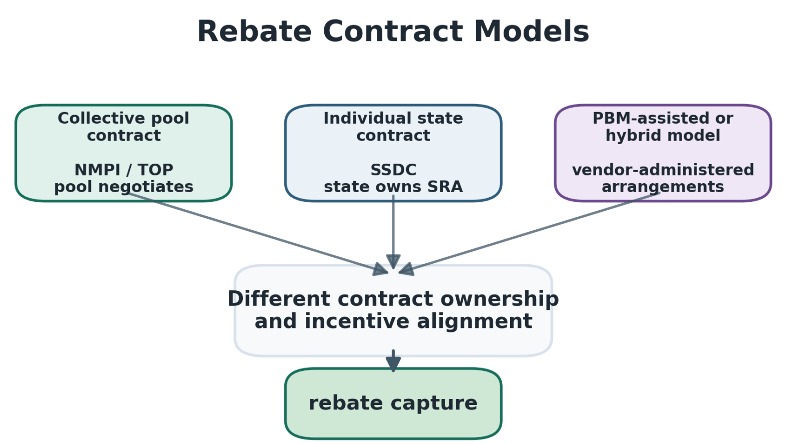

Figure 11: Rebate contract models

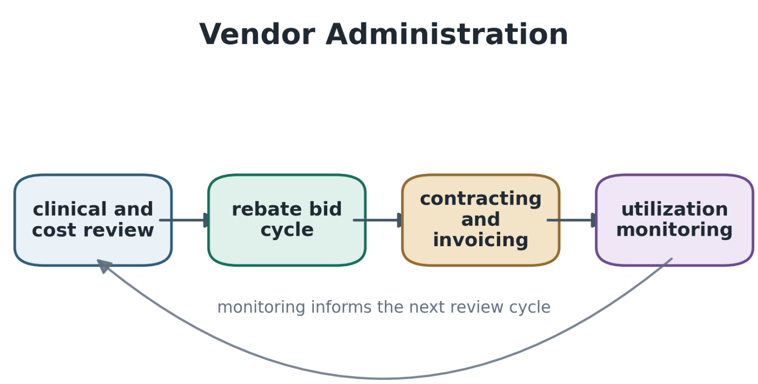

Figure 12: Vendor administration

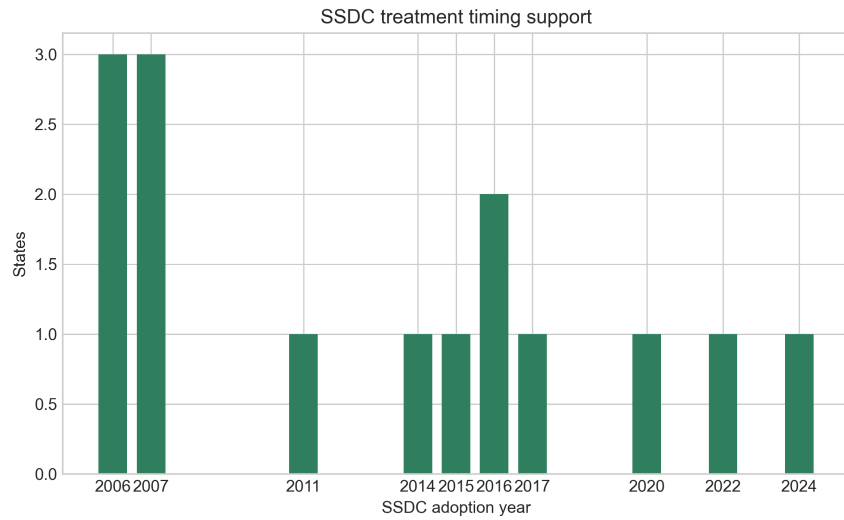

Figure 13: Appendix Figure G1. SSDC treatment timing support.

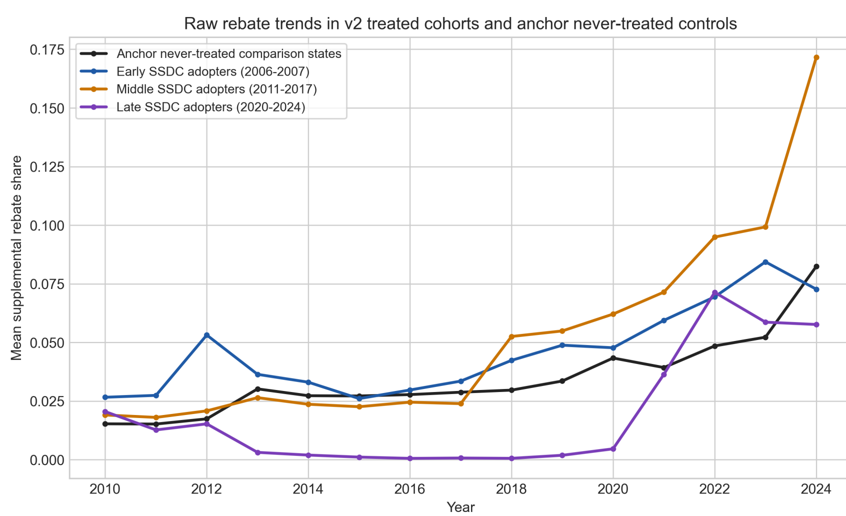

Figure 14: Appendix Figure G2. Raw rebate trends.

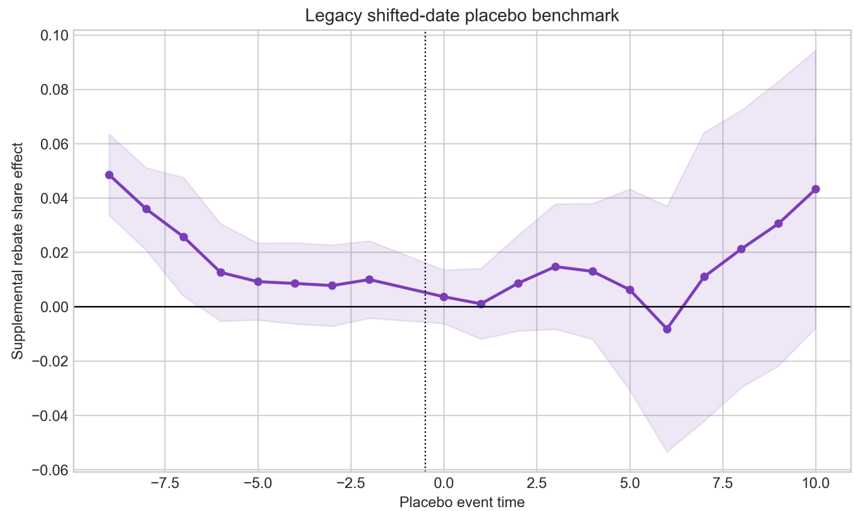

Figure 15: Appendix Figure G3. Shifted-date placebo.

Appendix I. Pool-Family Dynamic Extensions

The main text focuses on SSDC because it is the only pool family with enough later adopters to support a reasonably informative saturated dynamic design. To check whether that focus hides a stronger pattern elsewhere, I extend the later-adopter dynamic work across NMPI and TOP as an appendix exercise.

Pool family	Specification	Observations	States	Treated states	Average post effect
SSDC	Linear later-adopter event study + controls	619	43	9	0.0224
NMPI	Linear later-adopter event study + controls	504	35	2	-0.0095

Pool family	Specification	Observations	States	Treated states	Average post effect
TOP	Linear later-adopter event study + controls	589	41	3	-0.0053

Pool family	Saturated specification	Observations	States	Treated states	Average post effect	Note
SSDC	Uncontrolled	660	44	9	0.0337	Main paper estimate
SSDC	Controlled	649	44	9	0.0128	Main paper estimate
NMPI	Uncontrolled	525	35	2	-0.0158	Exploratory
NMPI	Controlled	504	35	2	-0.0158	Exploratory
TOP	Uncontrolled	630	42	3	-0.0133	Exploratory
TOP	Controlled	589	41	3	—	Saturated estimator failed

These extensions do not reveal a stronger positive dynamic pattern in NMPI or TOP. If anything, the broader later-adopter evidence is flat to negative outside SSDC, though support is thin enough that those family-level comparisons should remain secondary to the main SSDC design.

Appendix J. Pool-Family Interaction Extensions

I also estimate static interaction models that compare SSDC, NMPI, and TOP with no-pool state-years. The first version interacts pool family with `uniform_pdl`; the second adds pool-family-by-`pharmacy_carve_out` interactions.

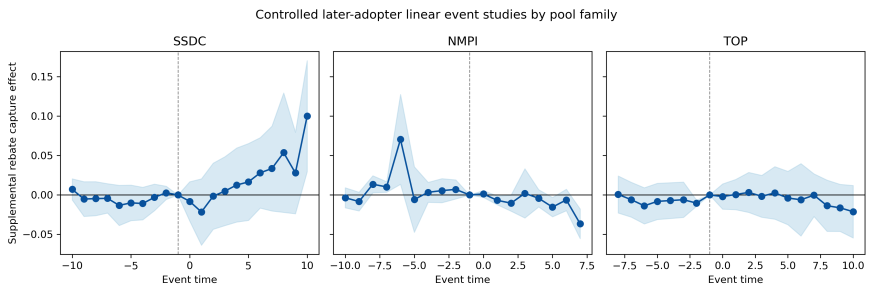

Figure 16: Appendix Figure I1. Pool-family event studies.

Pool family	Standalone pool effect vs no-pool	p-value	Pool x uniform PDL	p-value
SSDC	0.0091	0.658	0.0348	0.054
NMPI	-0.0374	0.008	0.0304	0.016
TOP	-0.0124	0.059	0.0284	0.054

Pool family	Pool x uniform PDL (with carve-out interactions)	p-value	Pool x carve-out	p-value
SSDC	0.0303	0.301	0.0060	0.821
NMPI	0.0286	0.176	-0.0030	0.879
TOP	0.0239	0.425	0.0079	0.790

The parsimonious model suggests that uniform-PDL settings look somewhat more favorable across all three pool families, not just SSDC. But once carve-out interactions are added, those uniform-PDL interaction estimates become imprecise, and the carve-out interactions themselves are all near zero. I do not interpret these static interaction models as evidence that carve-outs are a missing positive mechanism.

An SSDC-specific bundle extension points the same way. When I add an `ssdc_x_pharmacy_carve_out` term directly to the main bundle model, the interaction is 0.0127 with a standard error of 0.0185 and a `p` value of 0.494. The corresponding `ssdc_x_uniform_pdl` term is 0.0161 with a `p` value of 0.503. That direct carve-out check does not support a strong interaction story either.

Appendix K. Switchers from NMPI or TOP into SSDC

Five states move from NMPI or TOP into SSDC in the analytic file: Delaware, Kentucky, Pennsylvania, Vermont, and West Virginia. I compare those switchers with the 22 states that ever participate in NMPI or TOP but never join SSDC.

Specification	Comparison states	Observations	States	Estimate	Standard error	p-value
Switcher FE + uniform_pdl + pharmacy_carve_out	22	385	26	0.0315	0.0148	0.043
Switcher FE + full governance controls	22	385	26	0.0302	0.0211	0.164

The switcher result is suggestive but not stable enough to treat as a headline finding. The estimate stays positive after adding the fuller governance controls, but it is no longer statistically precise, and the switcher sample is small.

Appendix L. Federal-Rebate Headroom and Drug-Mix Checks

Two common concerns about this outcome are mechanical rather than causal. The first is that states with high federal rebate shares may have less room left for supplemental rebates, which could make supplemental rebate percentages look low even if bargaining is effective. The second is that states may simply use different drug baskets, so supplemental rebate percentages could reflect drug mix rather than negotiated leverage.

I address the first concern with two checks. One adds `national_rebate_pct` directly to the later-adopter linear benchmark and the bundle model. The other redefines the outcome as supplemental rebates divided by post-federal drug spending, `SuppRebate / (1 - national_rebate_pct)`.

Specification	Outcome	Observations	States	Average post effect / post_ssdc
Later-adopter linear benchmark + governance controls	SuppRebate	619	43	0.0224

Specification	Outcome	Observations	States	Average post effect / `post_ssdc`
Later-adopter linear benchmark + governance controls + national rebate share	`SuppRebate`	619	43	0.0201
Later-adopter linear benchmark + governance controls + national rebate share + SDUD mix controls	`SuppRebate`	619	43	0.0224
Later-adopter linear benchmark on supplemental rebate share of post-federal spending	`SuppRebate / (1 - national_rebate_pct)`	617	43	0.1860

Bundle specification	Outcome	Observations	States	`post_ssdc`	p-value
Baseline bundle model	`SuppRebate`	709	49	0.0127	0.571

Bundle specification	Outcome	Observations	States	`post_ssdc`	p-value
Bundle model + national rebate share	`SuppRebate`	709	49	0.0147	0.496
Bundle model + national rebate share + SDUD mix controls	`SuppRebate`	709	49	0.0154	0.476
Bundle model on supplemental rebate share of post-federal spending	`SuppRebate / (1 - national_rebate_pct)`	706	49	0.0460	0.544

These checks do not support the headroom concern as the main explanation for the paper's result. In the full panel, `SuppRebate` and `national_rebate_pct` are positively correlated rather than negatively offsetting one another, and conditioning on national rebate intensity does not recover a meaningful standalone SSDC effect.

I address the second concern with SDUD-based state-year utilization measures. The SDUD panel covers 1,142 state-years in the merged file and captures drug mix through rebate-drug category shares, average spend per prescription, and NDC-level concentration. The category crosswalk is imperfect in early years but becomes much stronger over time: matched spending rises from 26.6 percent in 2002 to 99.5 percent in 2024, with an average of 79.4 percent across 2010-2024.

Pool family	Single-source spend share	Innovator-multiple-source spend share	Avg spend per prescription	Top-10 NDC spend share
NoPool	0.2447	0.2415	79.72	0.1421
SSDC	0.4546	0.2327	91.46	0.1868
NMPI	0.3460	0.2387	88.48	0.1637
TOP	0.3519	0.2709	88.57	0.1581

Those descriptive differences confirm that state drug baskets do vary. But they do not rescue a positive pool-only interpretation. Once I add the SDUD mix controls to the bundle model, the `post_ssdc` coefficient remains small and imprecise at 0.0154 (`p = 0.476`). In that same model, `national_rebate_pct`, single-source spend share, and average spend per prescription are all positively associated with supplemental rebate capture. So the paper's main null is not just proxying for cross-state differences in drug usage mix.

Appendix M. Exploratory Pool-Size Heterogeneity

One possibility is that pooling matters mainly for smaller Medicaid programs, which may gain the most from borrowing larger members' bargaining leverage. I evaluate that idea in a separate size-heterogeneity branch using pre-entry Medicaid enrollment.

Pool family	Relative-size interaction	p-value	Absolute-size interaction	p-value	Interpretation
SSDC	0.0170	0.008	-0.0031	0.634	Positive only in the relative-size spec
NMPI	0.0006	0.922	-0.0054	0.392	Null
TOP	0.0102	0.804	0.0048	0.598	Null

The SSDC relative-size slope is not strong enough to change the paper's main interpretation. It survives dropping Kentucky, but it weakens sharply when Ohio is excluded and disappears when the SSDC treated sample is restricted to entrants that were actually smaller than the largest incumbent pool member. Taken together, the current evidence does not show a robust across-pool pattern in which smaller Medicaid programs systematically benefit more from joining a pool.

Appendix N. Any-Pool Specification and Triple Interactions

As a complement to the pool-family-specific analyses, I estimate models using an `in_any_pool` indicator that collapses SSDC, NMPI, and TOP into a single binary variable. This directly tests whether pooling *in general*—regardless of which pool—increases supplemental rebate capture.

Model	Term	Estimate	SE	p-value
(1) Any-pool standalone	Any pool	0.0006	0.0098	0.951
(2) + governance	Any pool	0.0007	0.0102	0.946
(3) + any-pool × PDL	Any pool × uniform PDL	0.0231	0.0128	0.078
(4) + carve-out interaction	Any pool × uniform PDL	0.0069	0.0178	0.700
	Any pool × carve-out	0.0256	0.0214	0.241
(5) Triple interaction	Any pool × PDL × carve-out	0.0891	0.0299	0.005

All models: state + year FE, state-clustered SEs, N = 709, 49 states. Model (5) drops the two-way uniform PDL × carve-out interaction due to collinearity.

The standalone any-pool coefficient is effectively zero across all specifications, confirming that pooling by itself does not raise supplemental rebate capture. The triple interaction (8.91 pp, p = 0.005) captures states that simultaneously participate in a pool, maintain a uniform PDL, and have carved the pharmacy benefit out of managed care. This is the only specification in the full paper that produces a statistically significant positive any-pool-related coefficient.

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